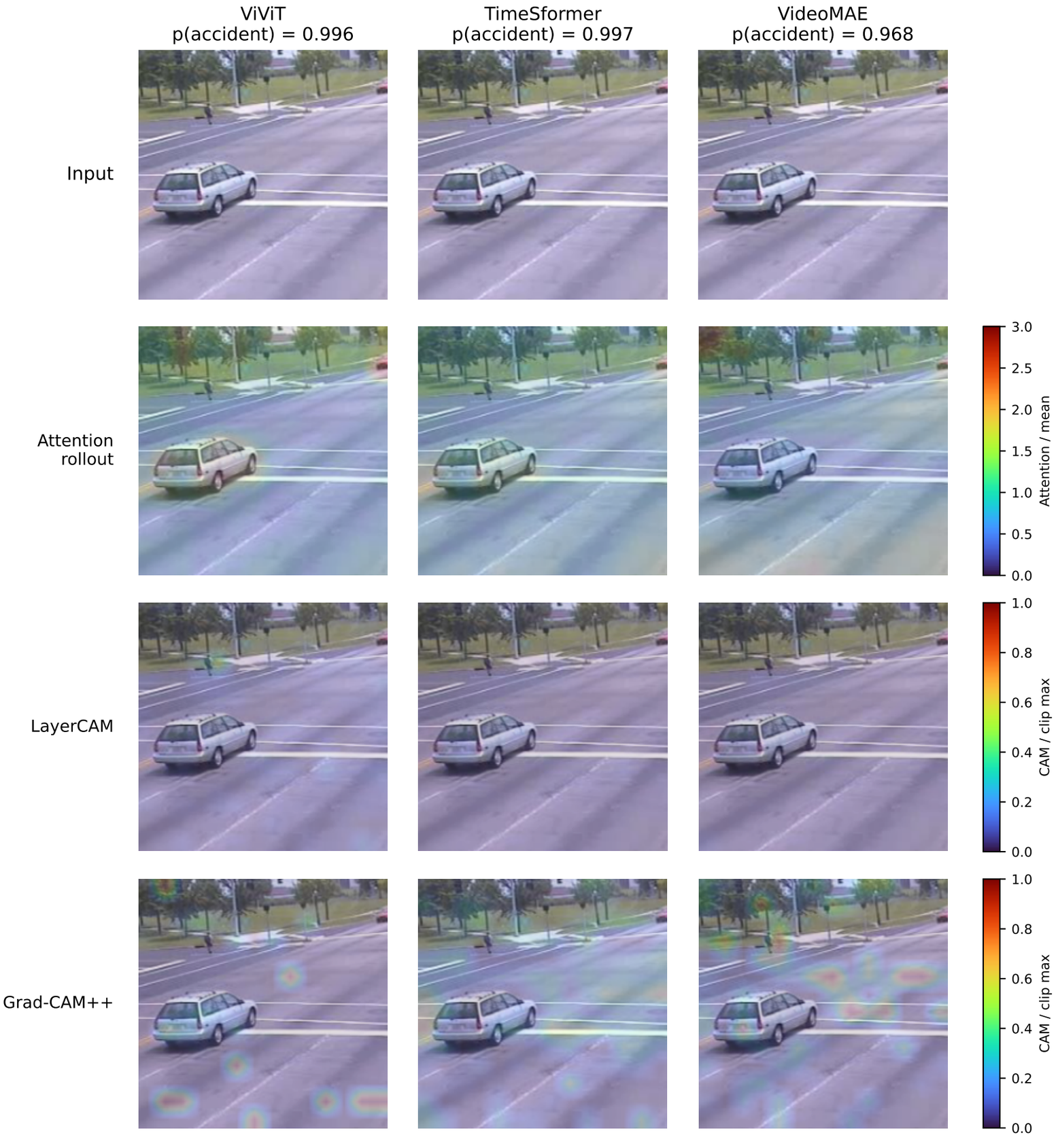
A01 | Accident (dataset label) | t = 8.44 s

Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A01 | Attention rollout | Accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

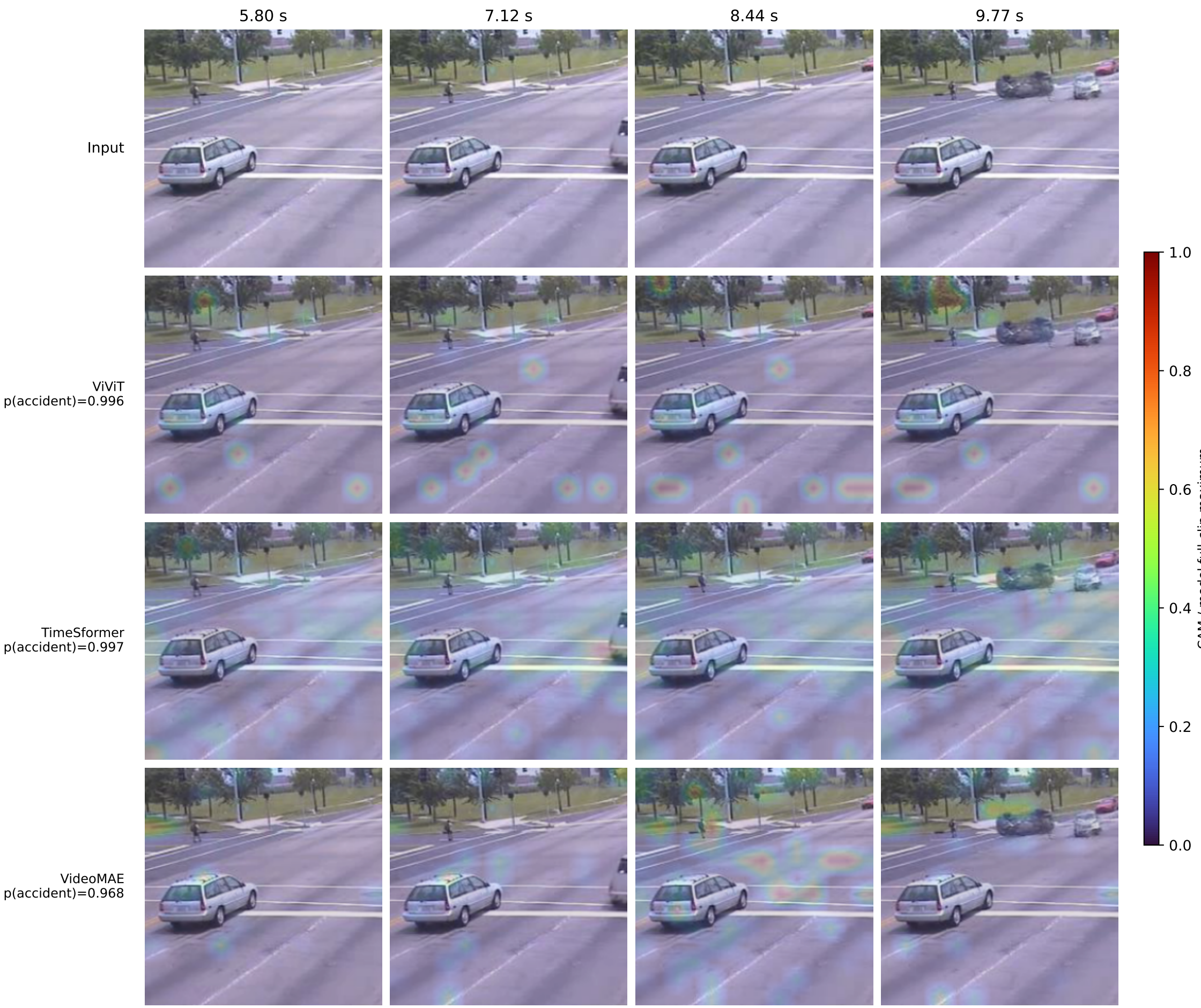
A01 | LayerCAM | Accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A01 | Grad-CAM++ | Accident label

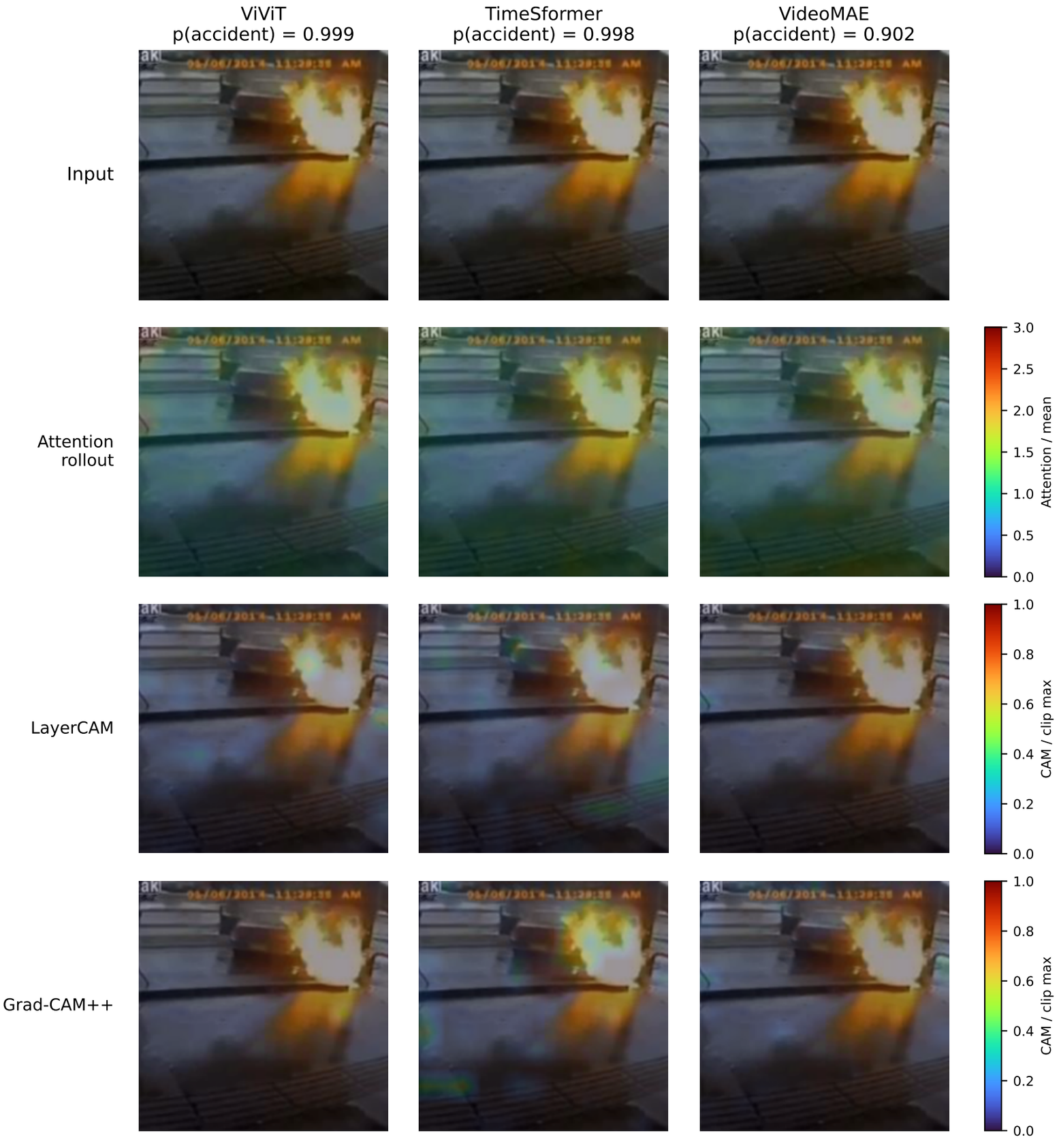
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A02 | Accident (dataset label) | t = 29.46 s

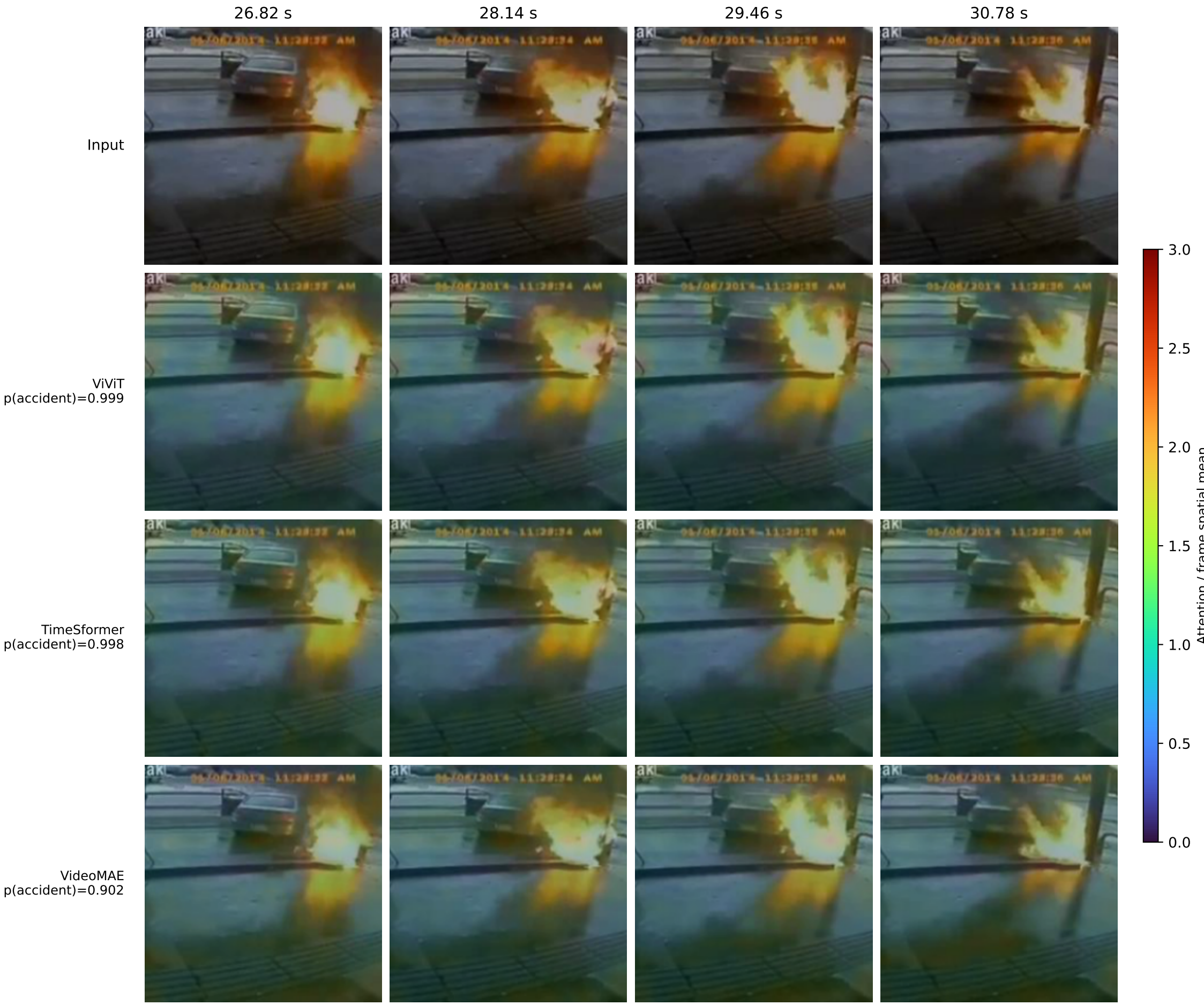
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A02 | Attention rollout | Accident label

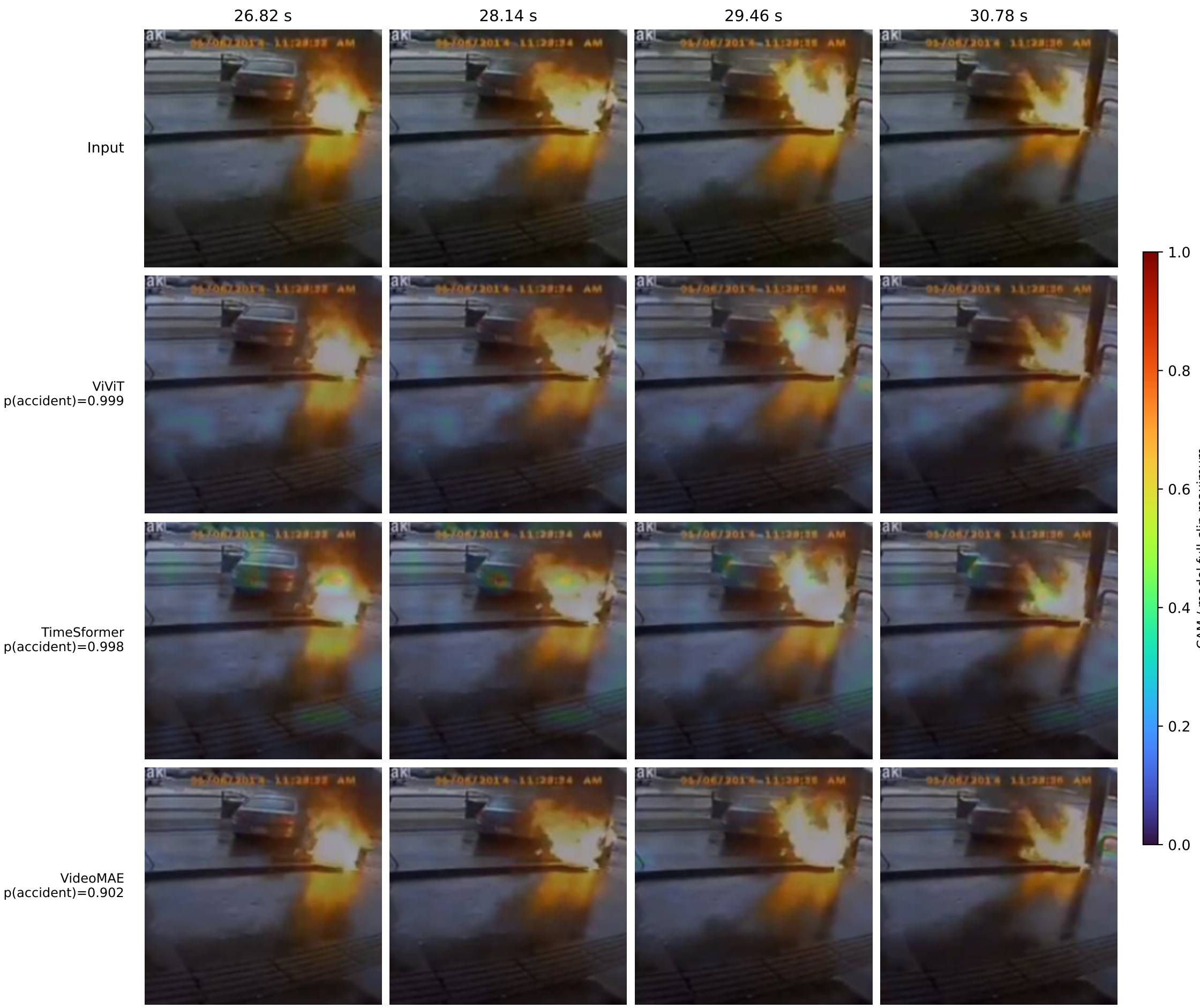
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A02 | LayerCAM | Accident label

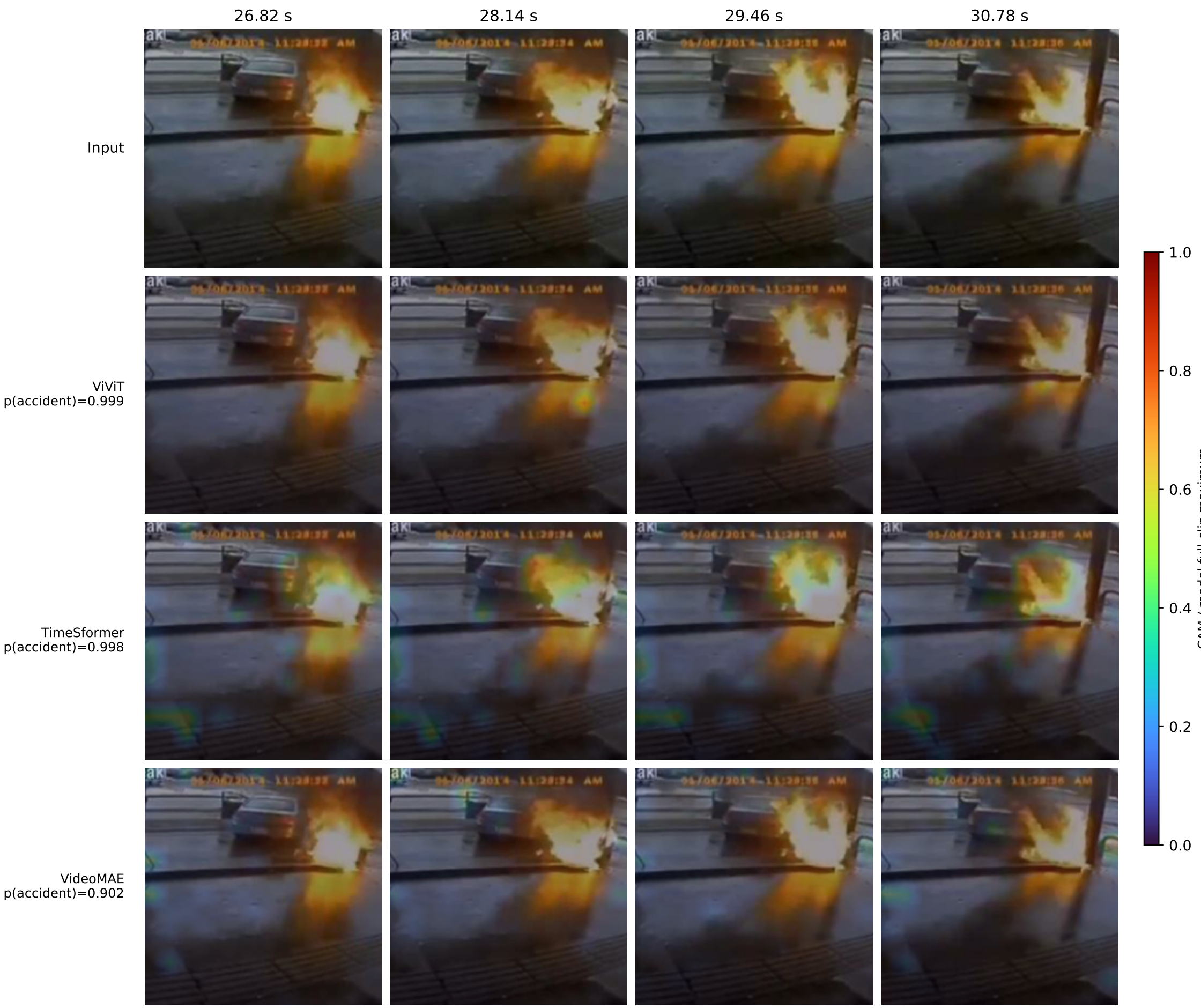
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A02 | Grad-CAM++ | Accident label

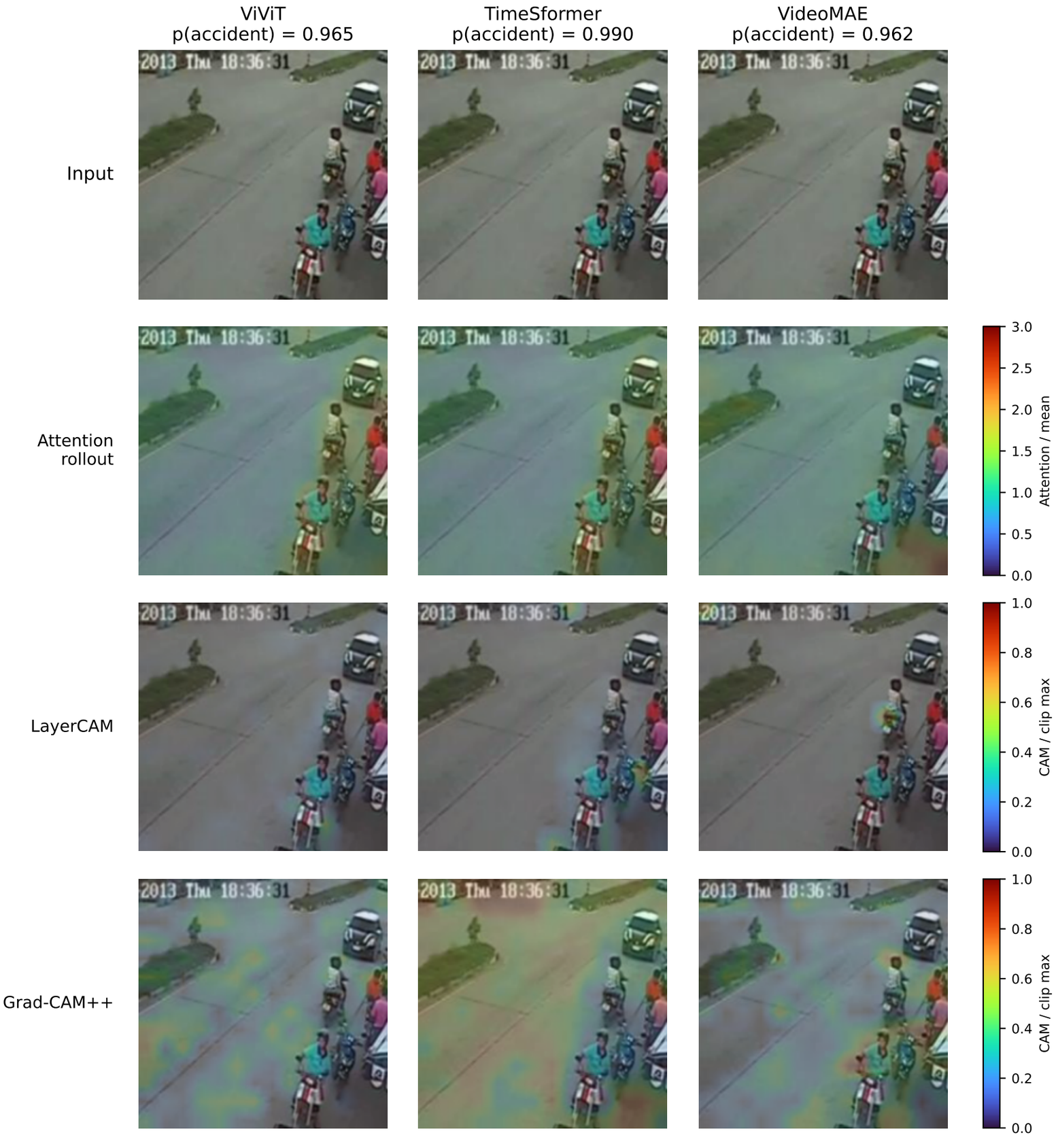
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A03 | Accident (dataset label) | t = 22.54 s

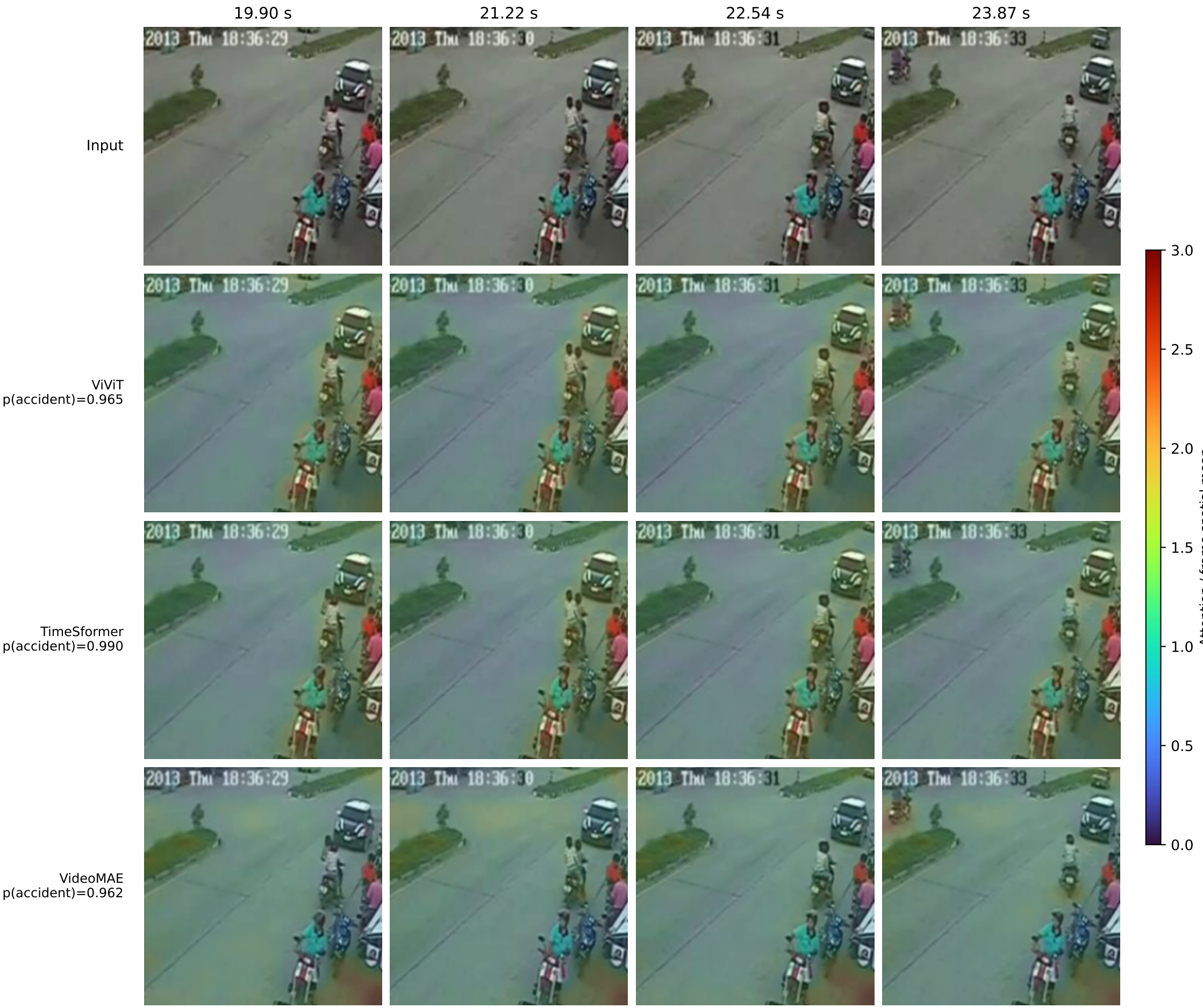
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A03 | Attention rollout | Accident label

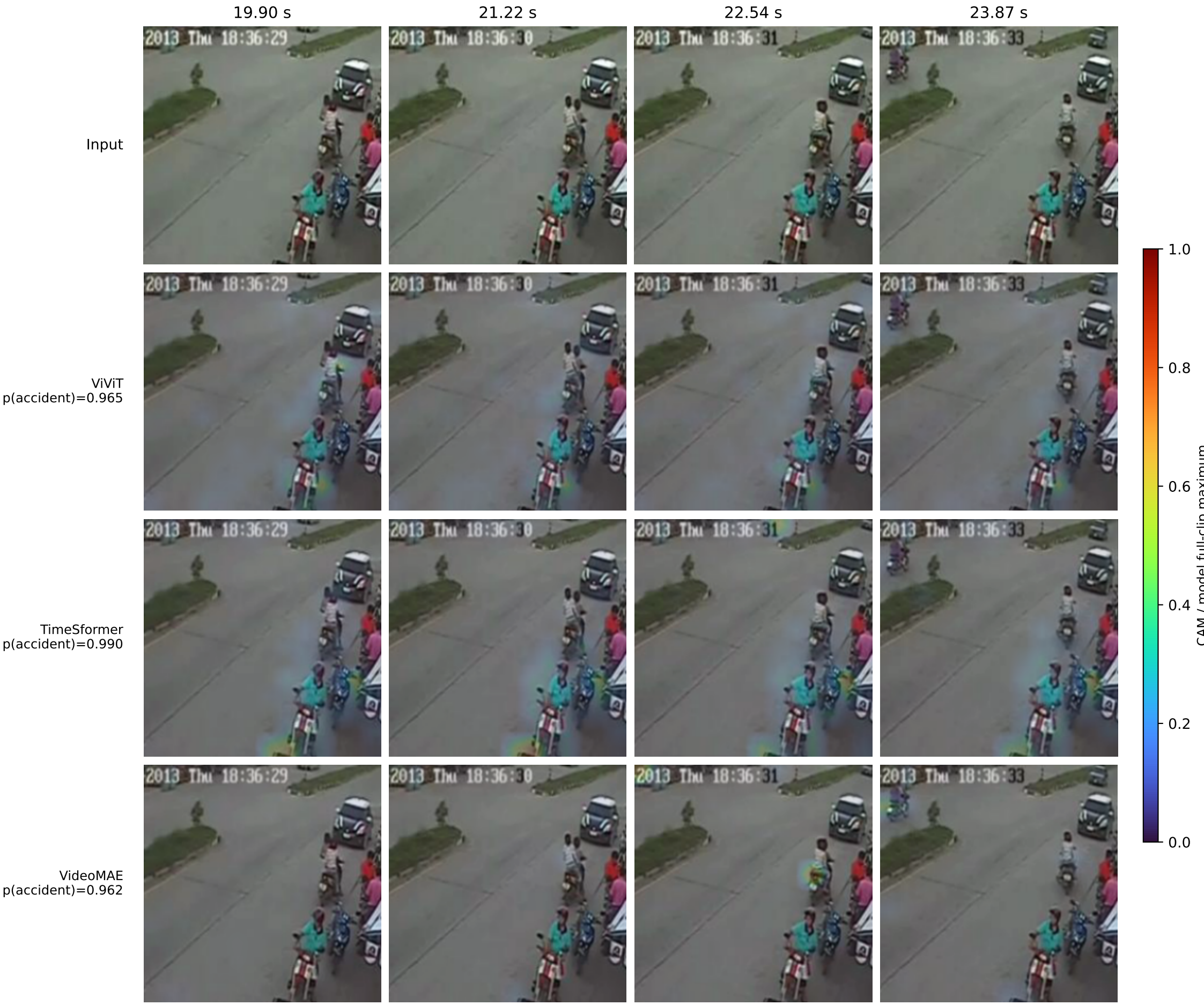
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A03 | LayerCAM | Accident label

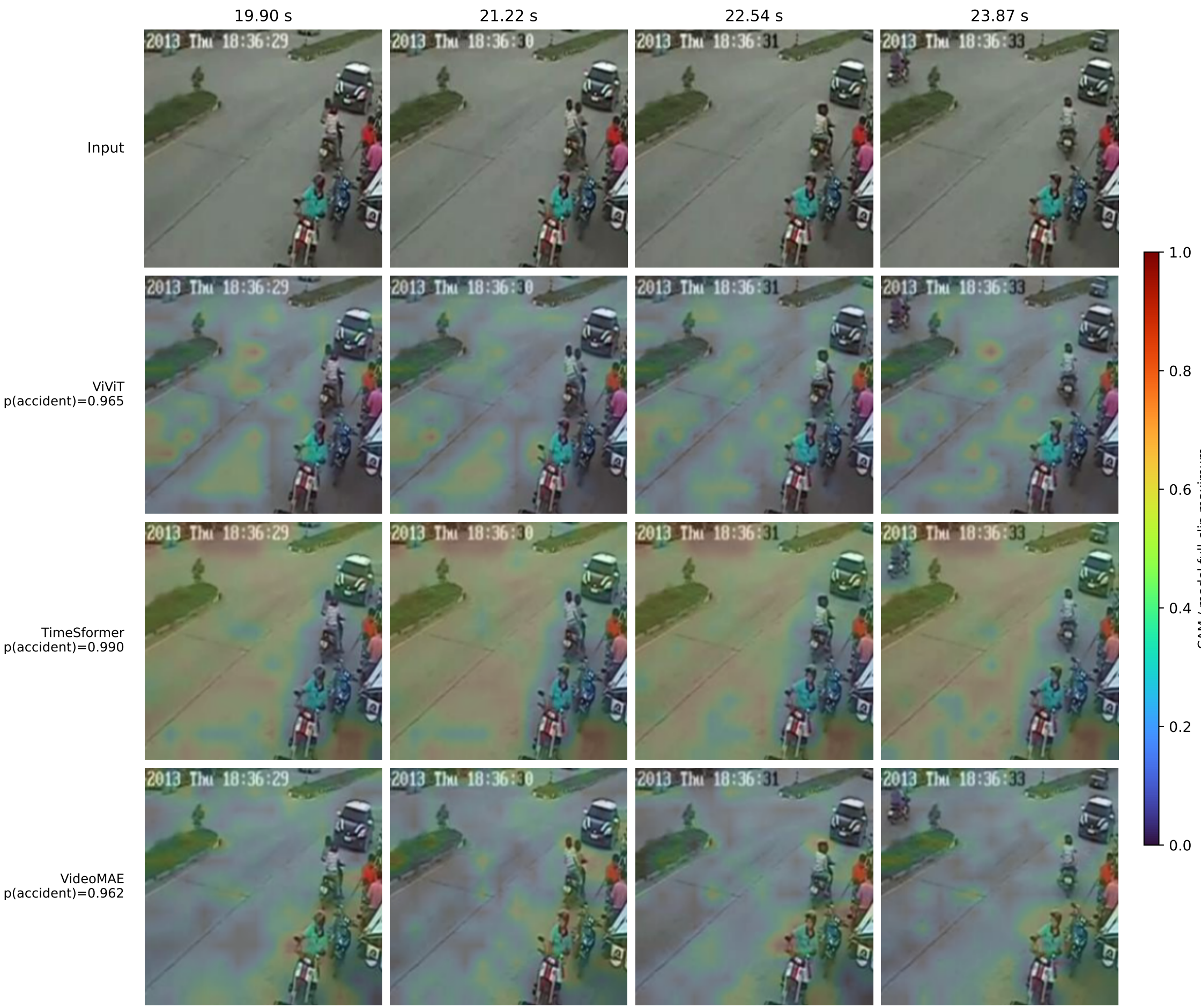
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A03 | Grad-CAM++ | Accident label

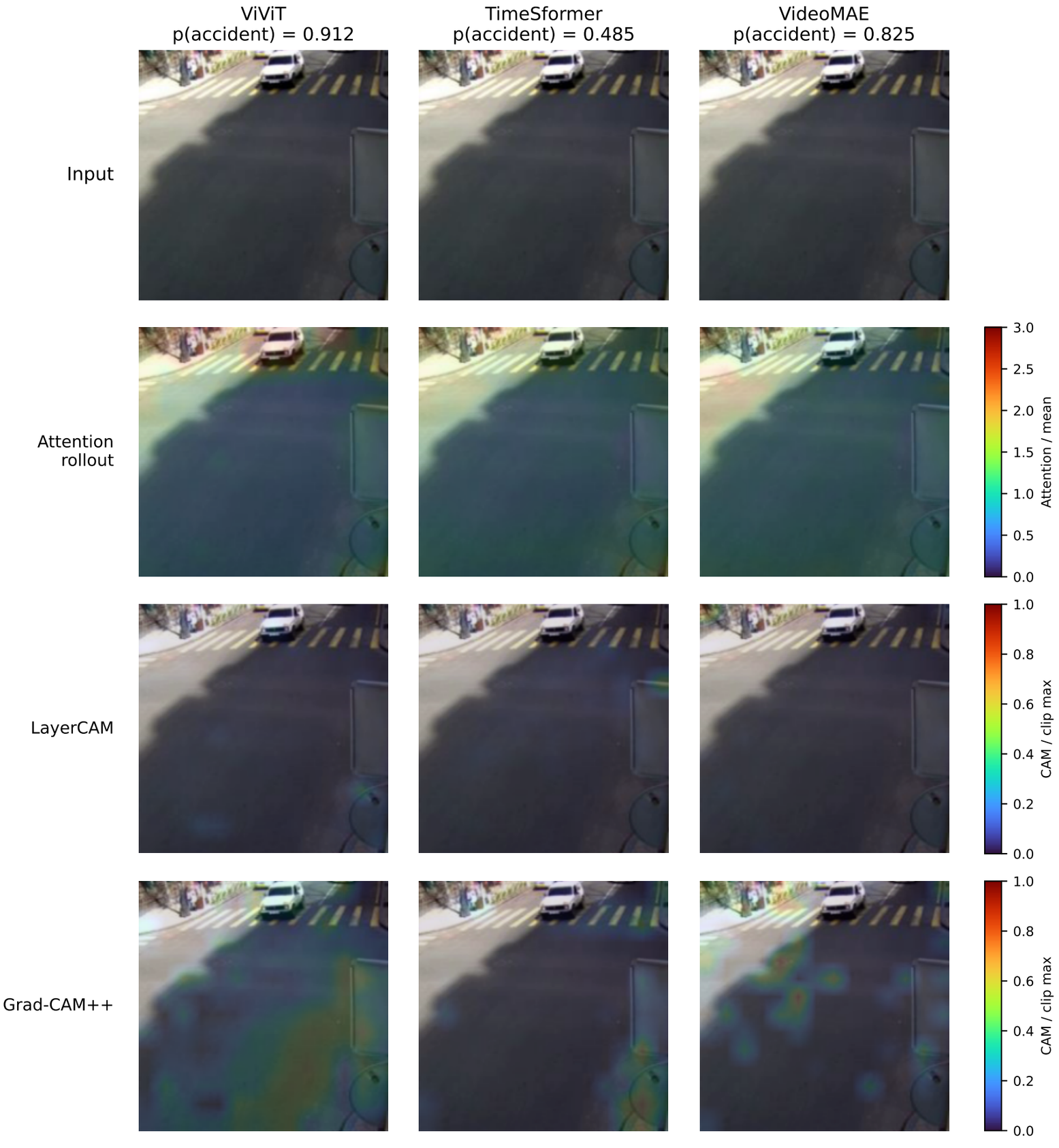
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A04 | Accident (dataset label) | t = 2.29 s

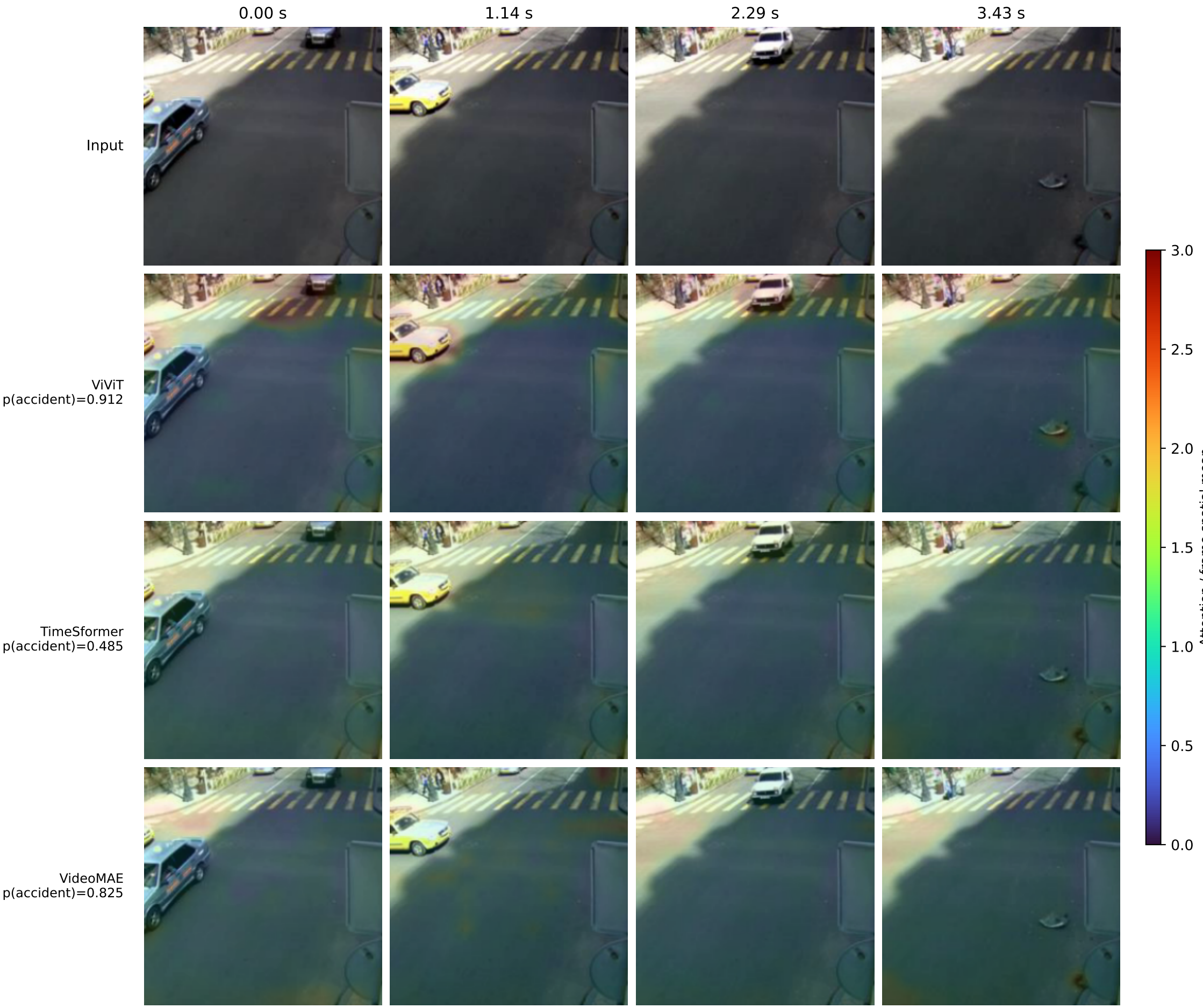
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A04 | Attention rollout | Accident label

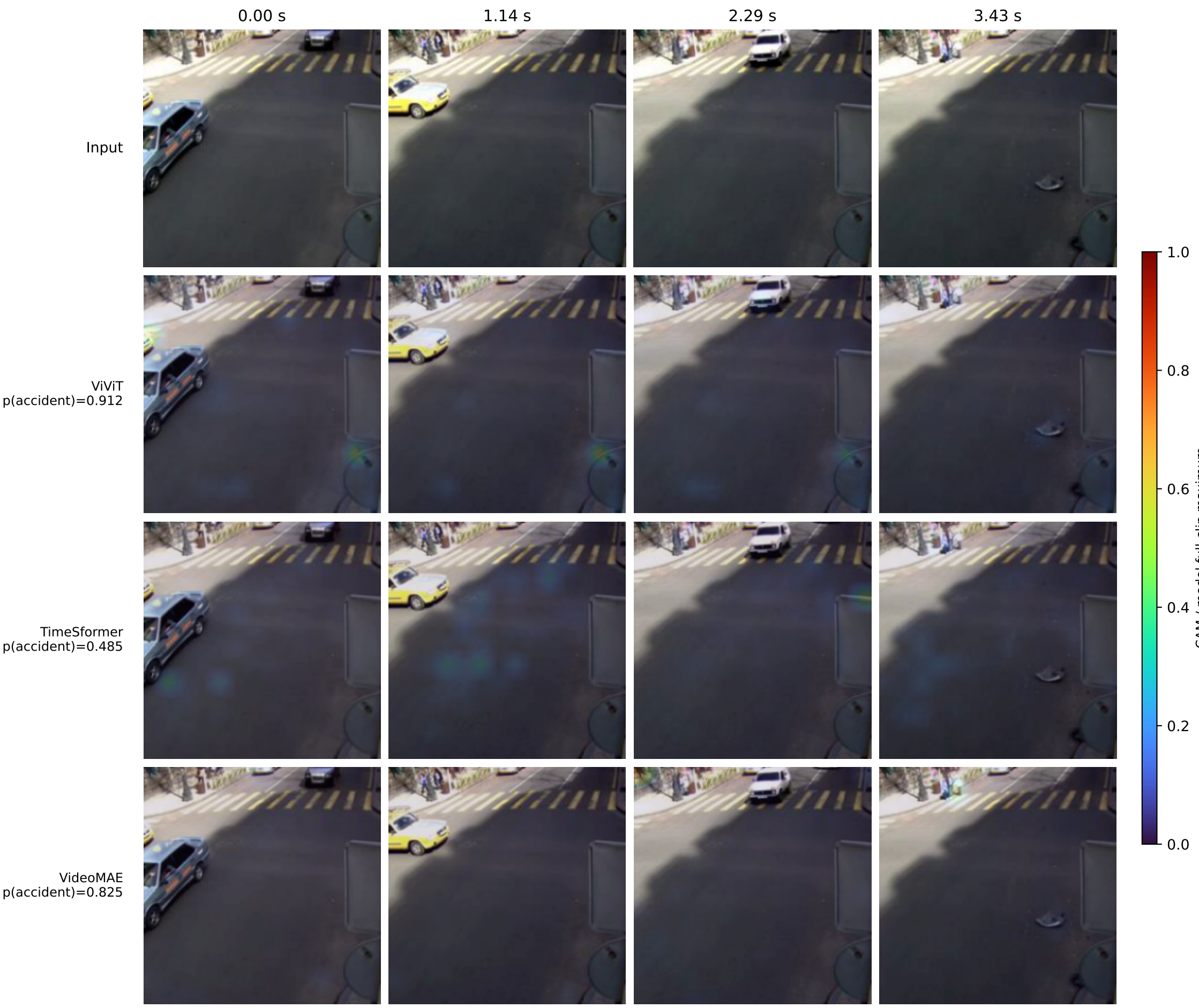
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A04 | LayerCAM | Accident label

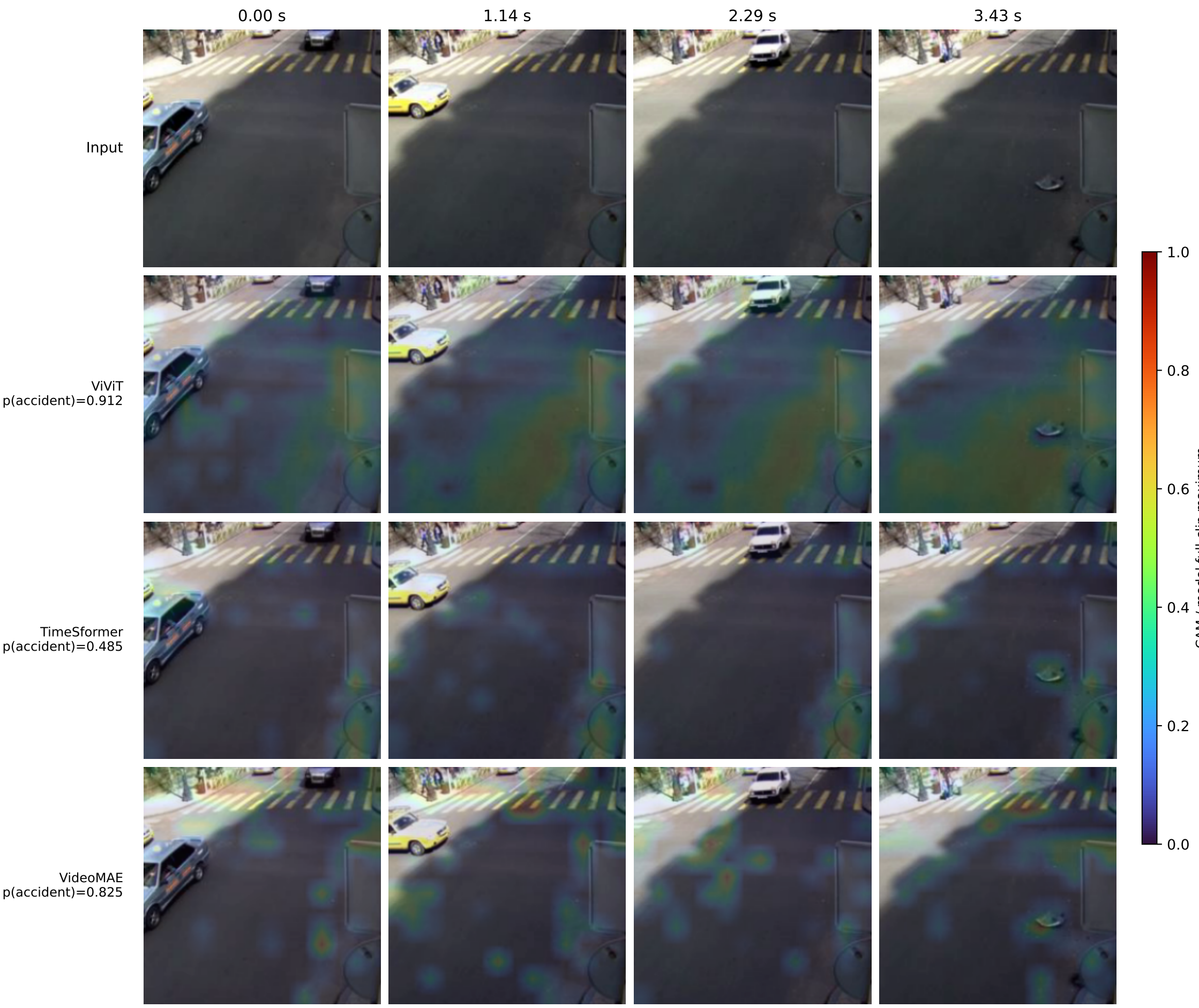
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A04 | Grad-CAM++ | Accident label

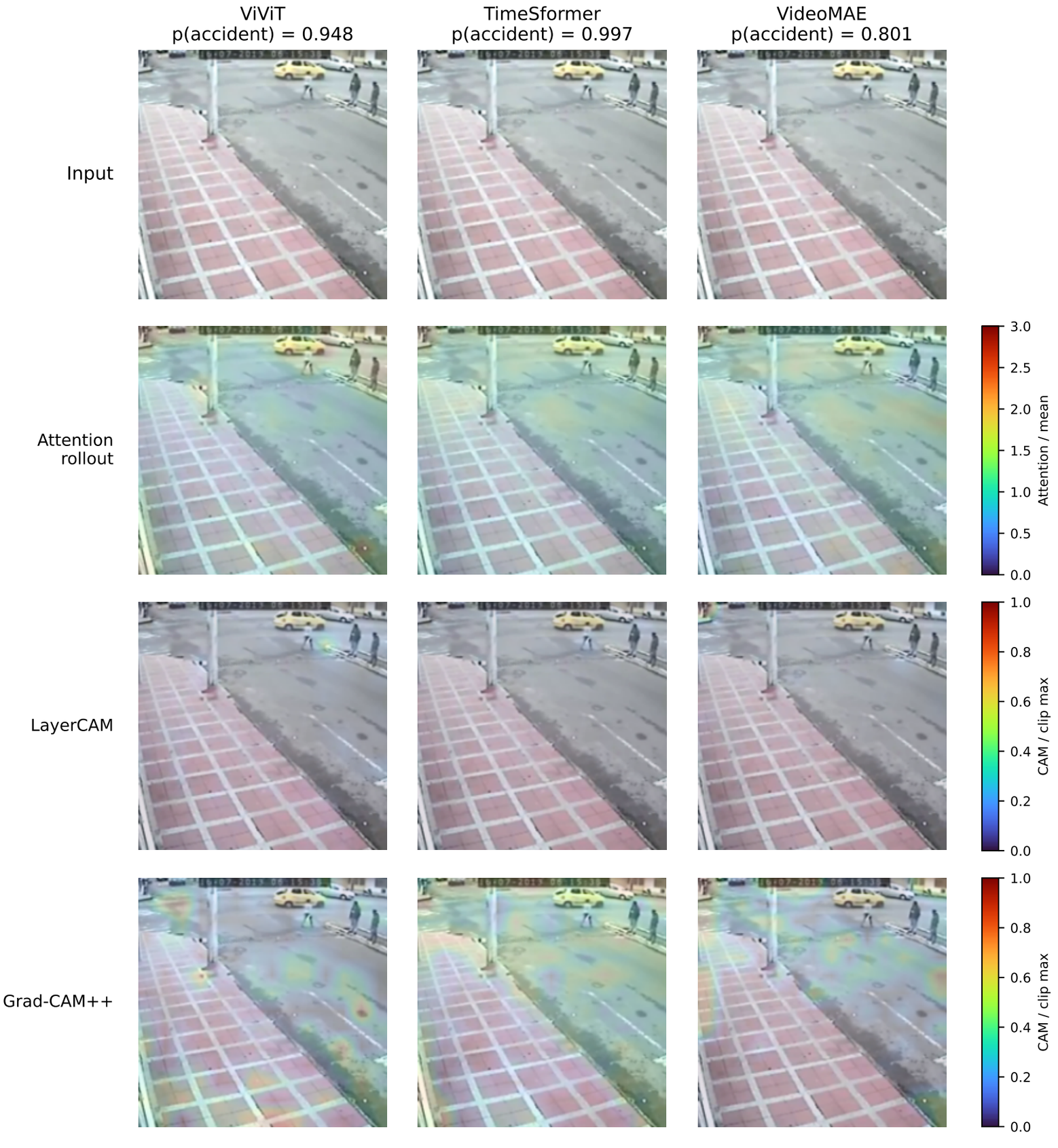
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A05 | Accident (dataset label) | t = 22.14 s

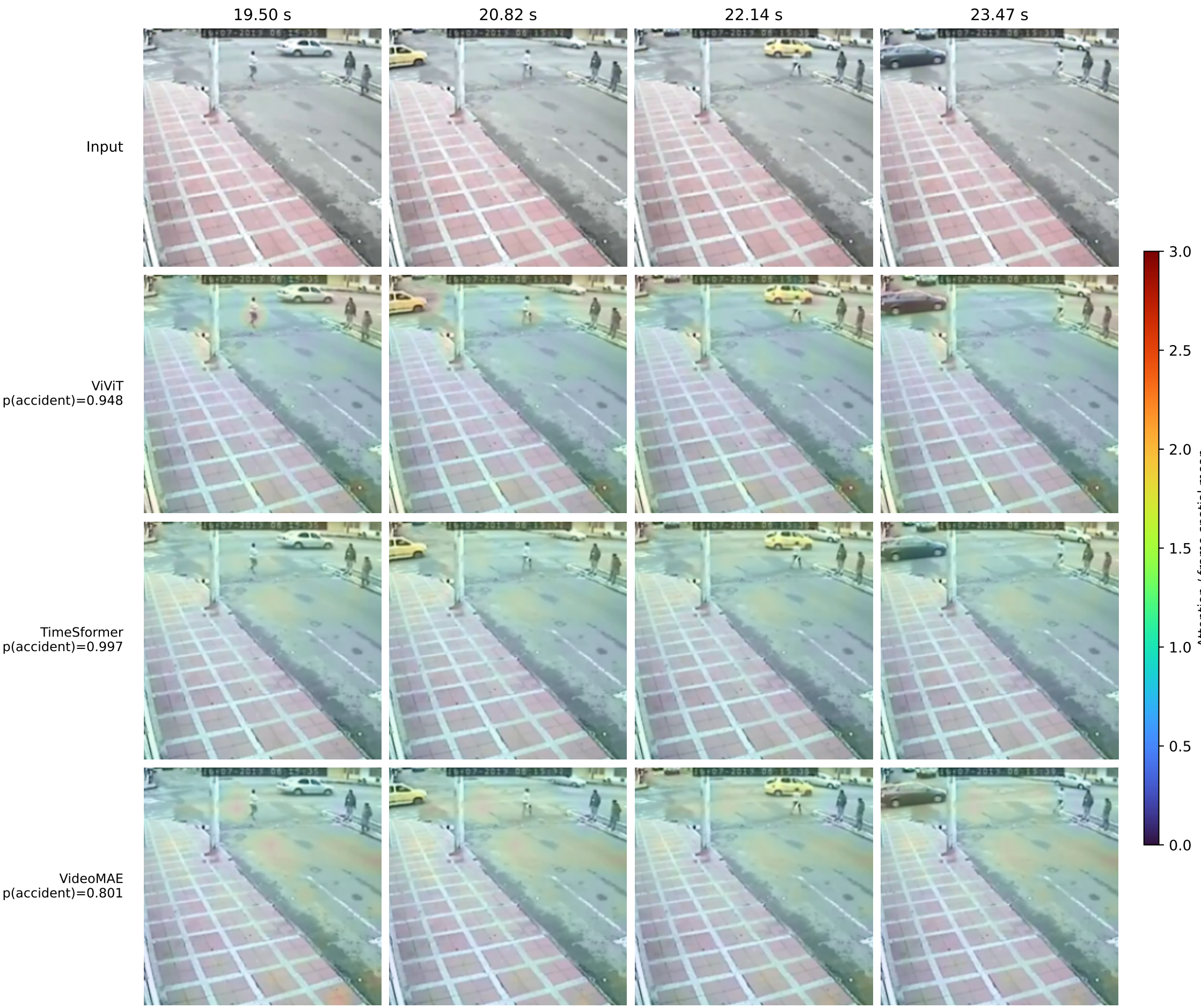
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A05 | Attention rollout | Accident label

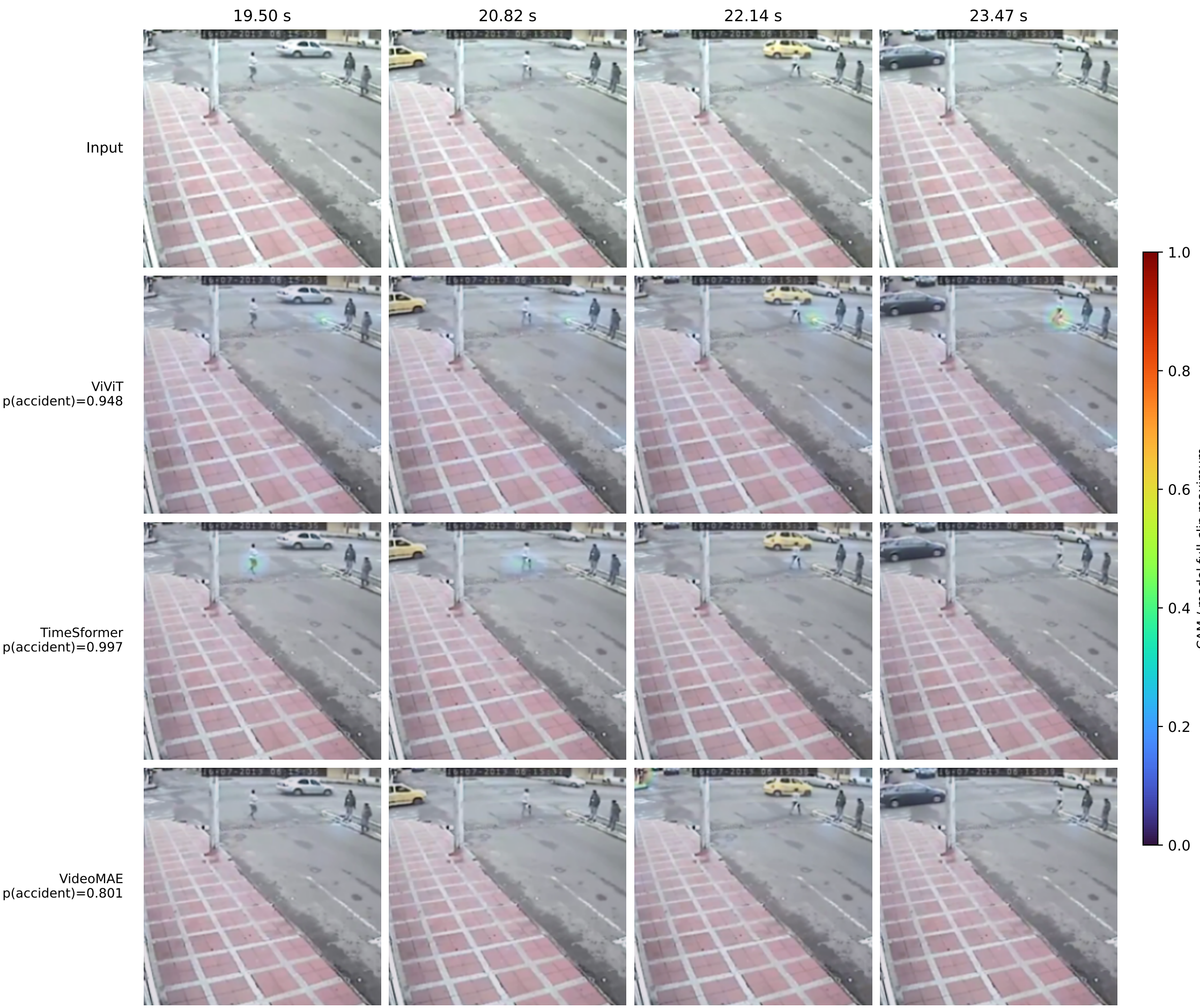
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A05 | LayerCAM | Accident label

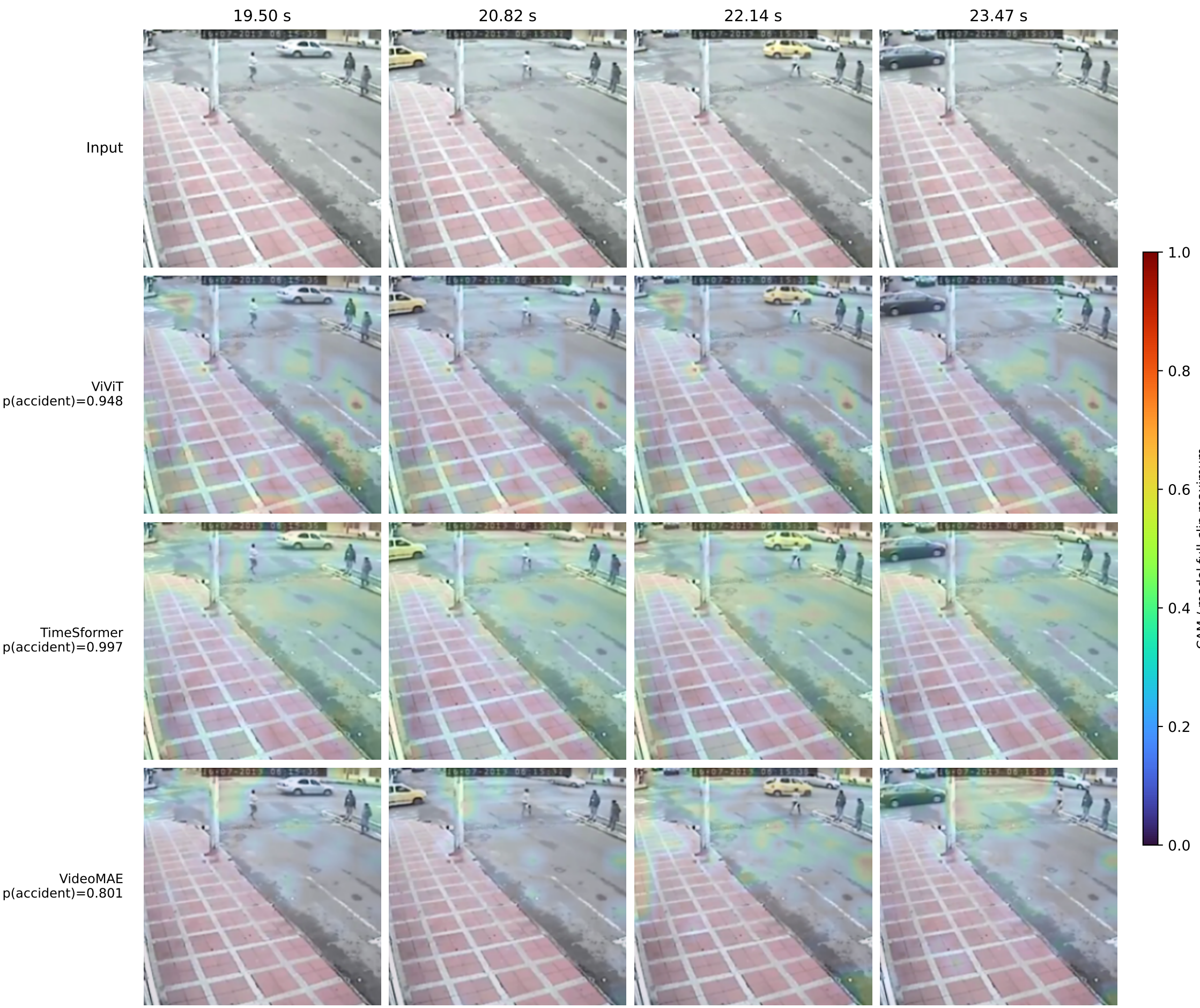
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A05 | Grad-CAM++ | Accident label

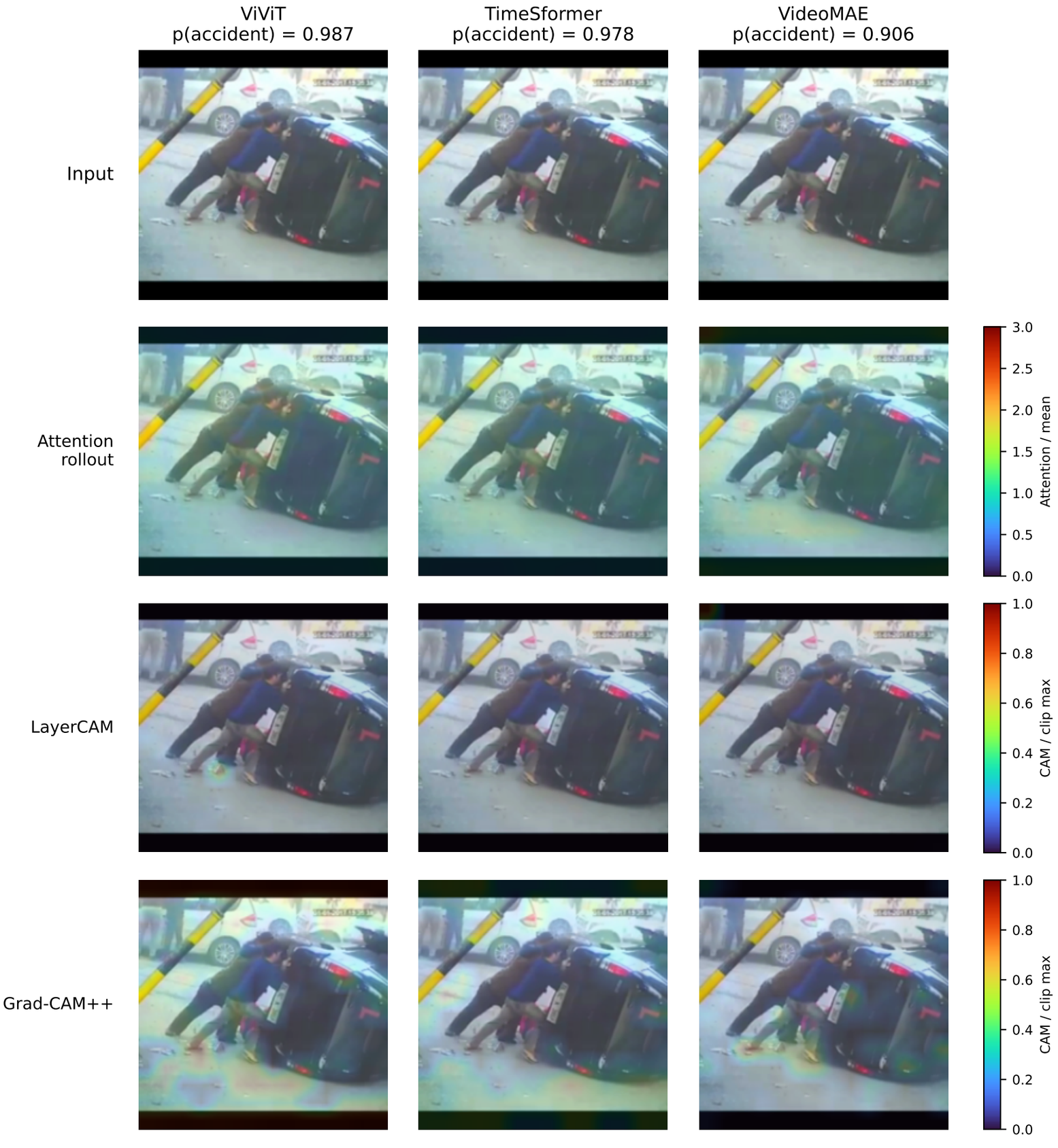
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A06 | Accident (dataset label) | t = 38.33 s

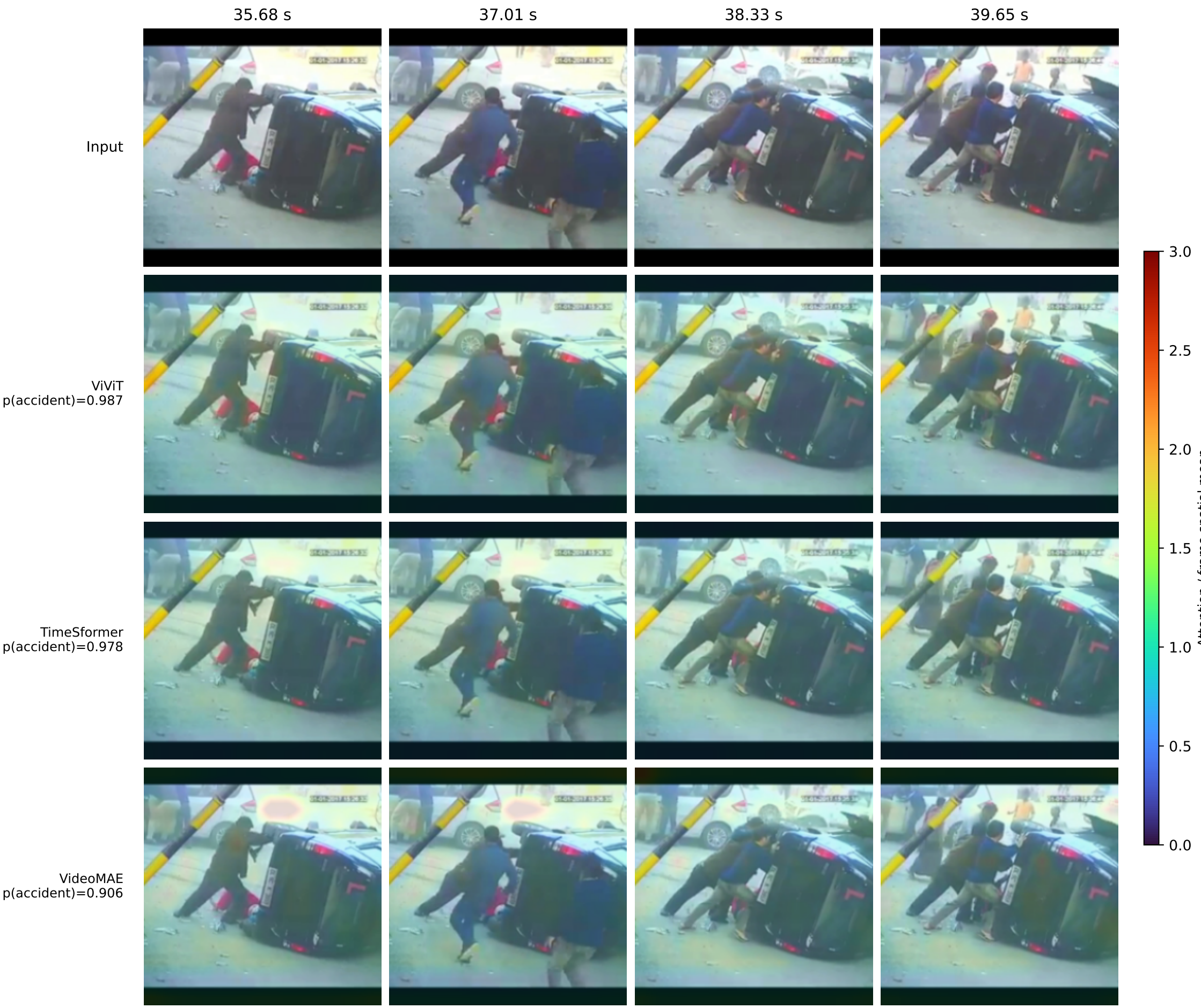
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A06 | Attention rollout | Accident label

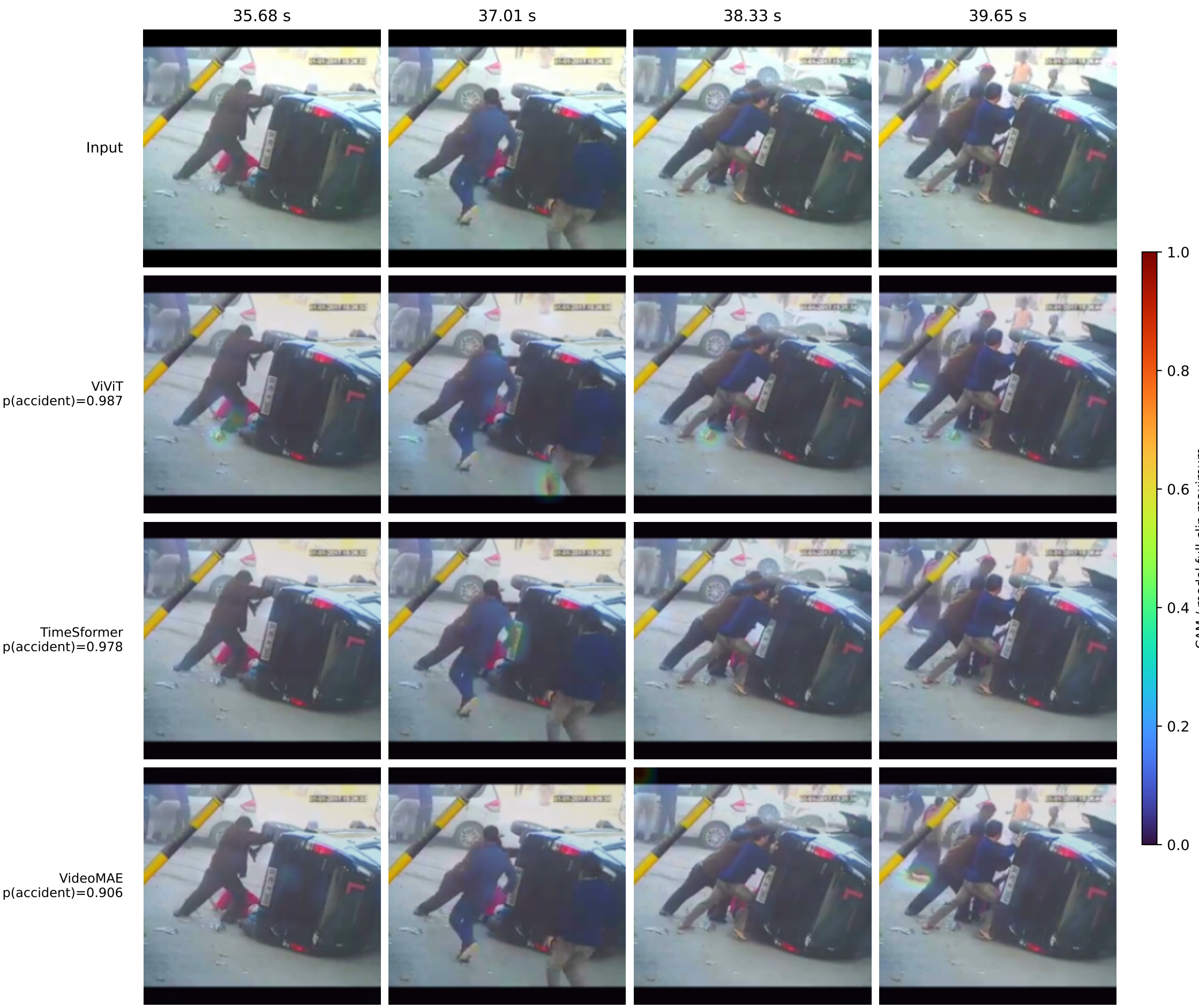
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A06 | LayerCAM | Accident label

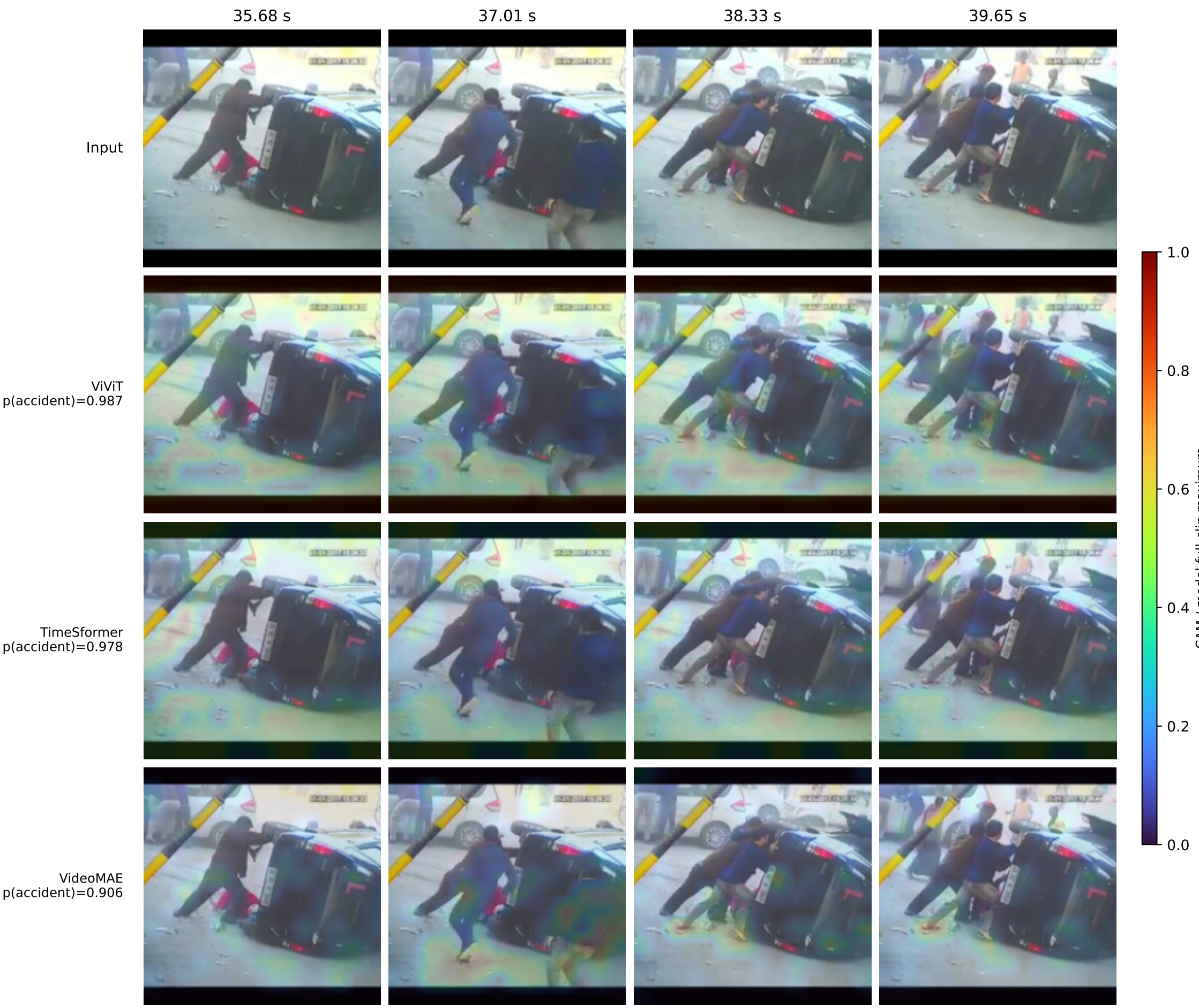
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A06 | Grad-CAM++ | Accident label

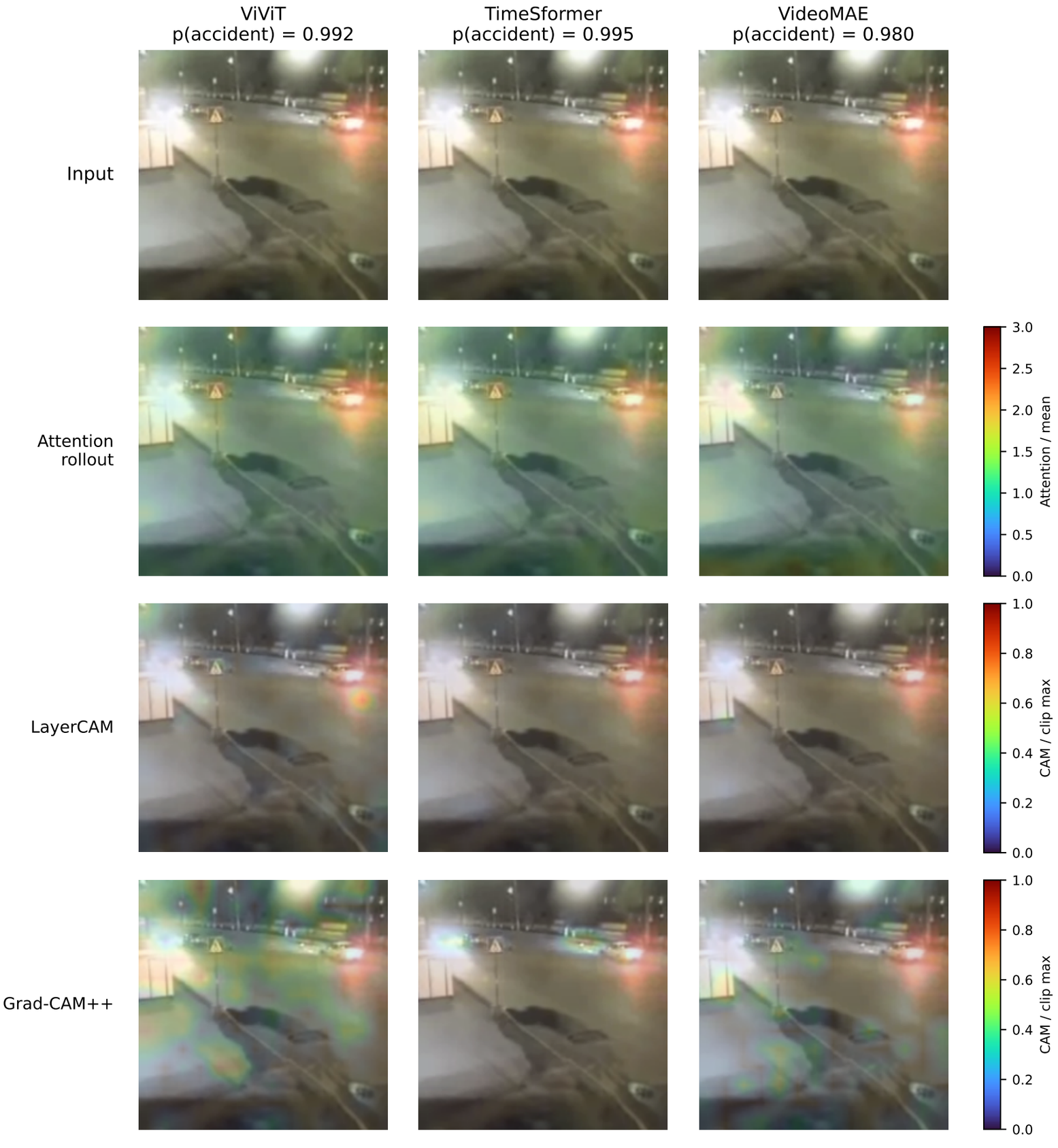
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A07 | Accident (dataset label) | t = 10.43 s

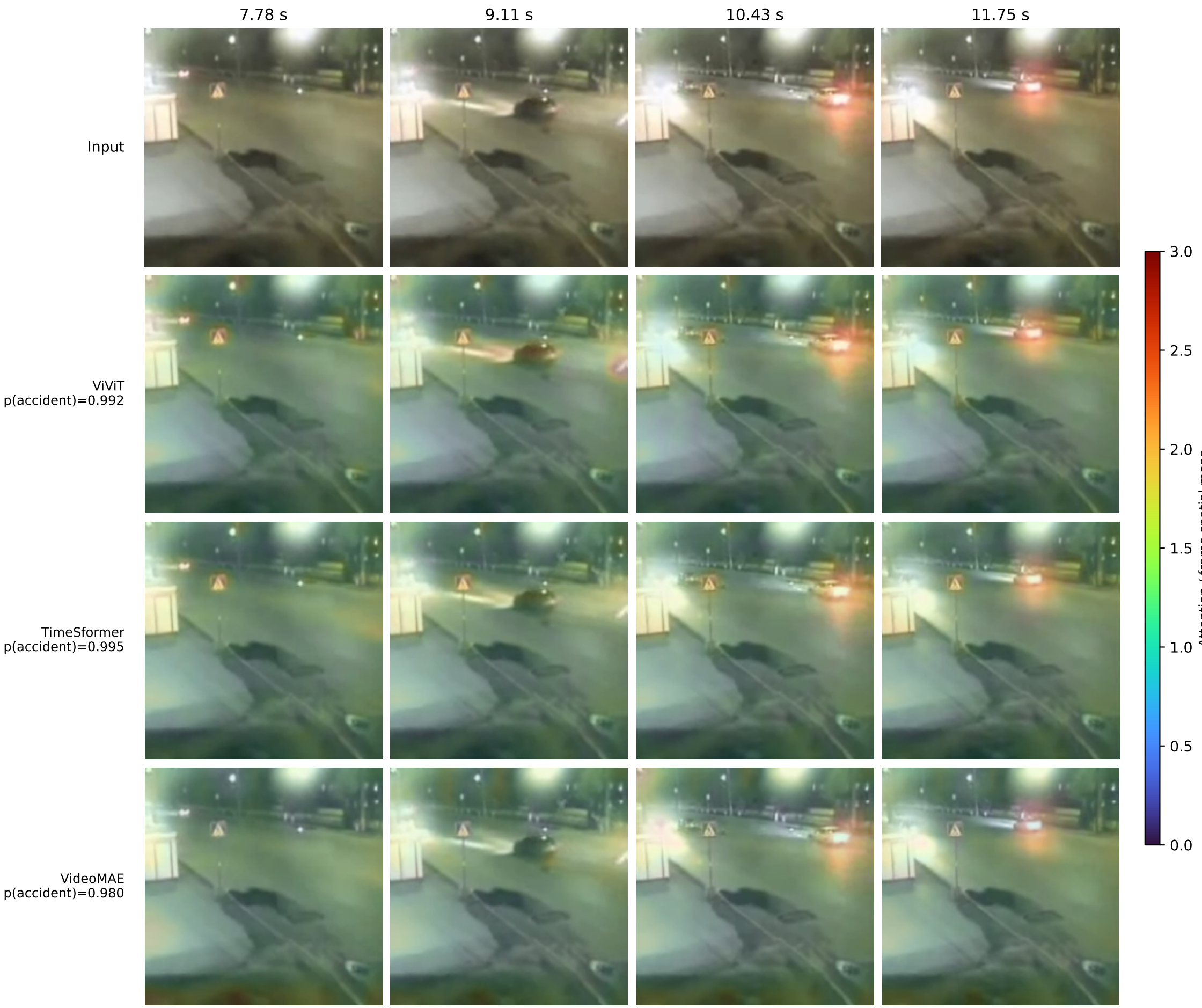
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A07 | Attention rollout | Accident label

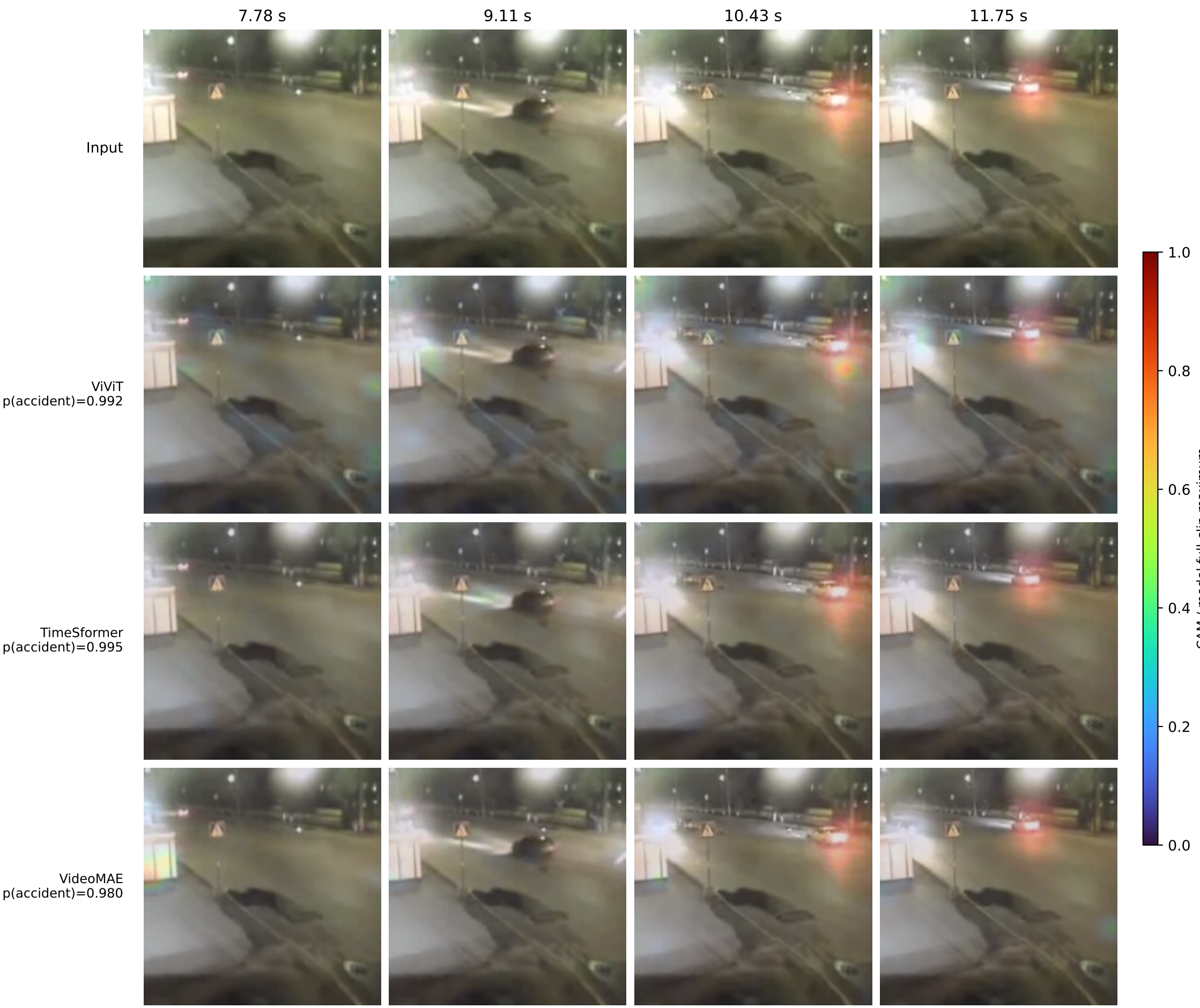
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A07 | LayerCAM | Accident label

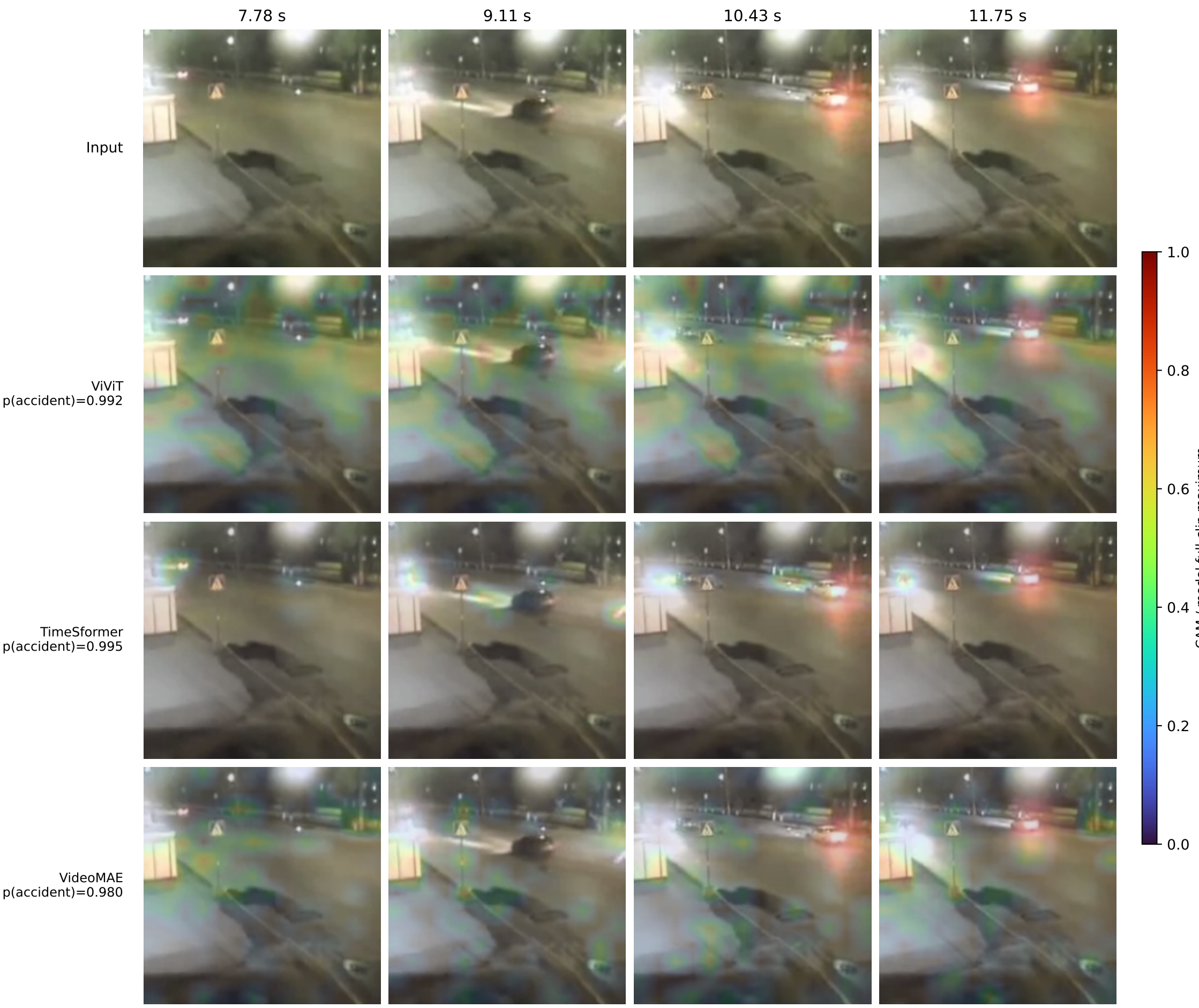
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A07 | Grad-CAM++ | Accident label

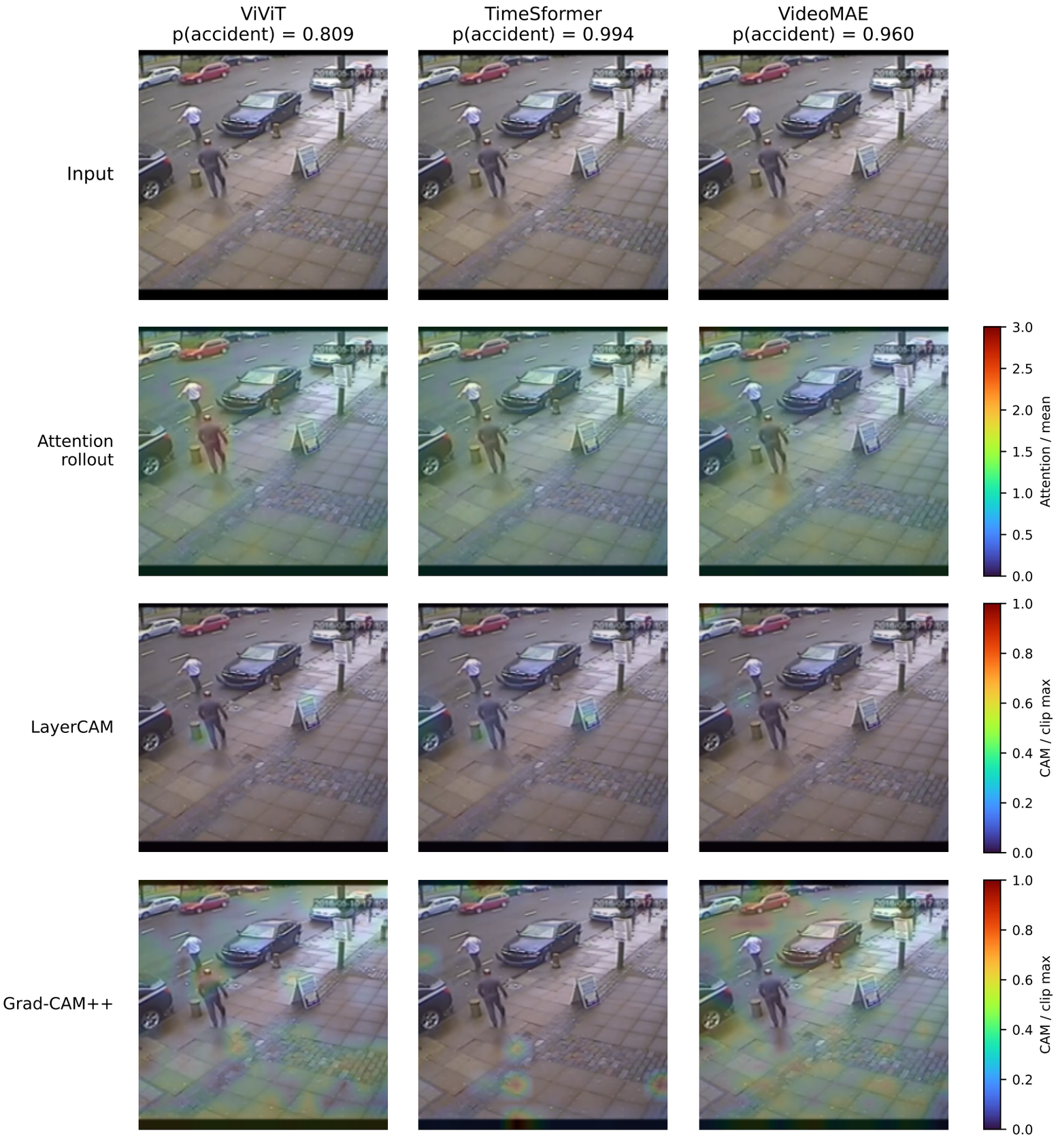
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A08 | Accident (dataset label) | t = 29.63 s

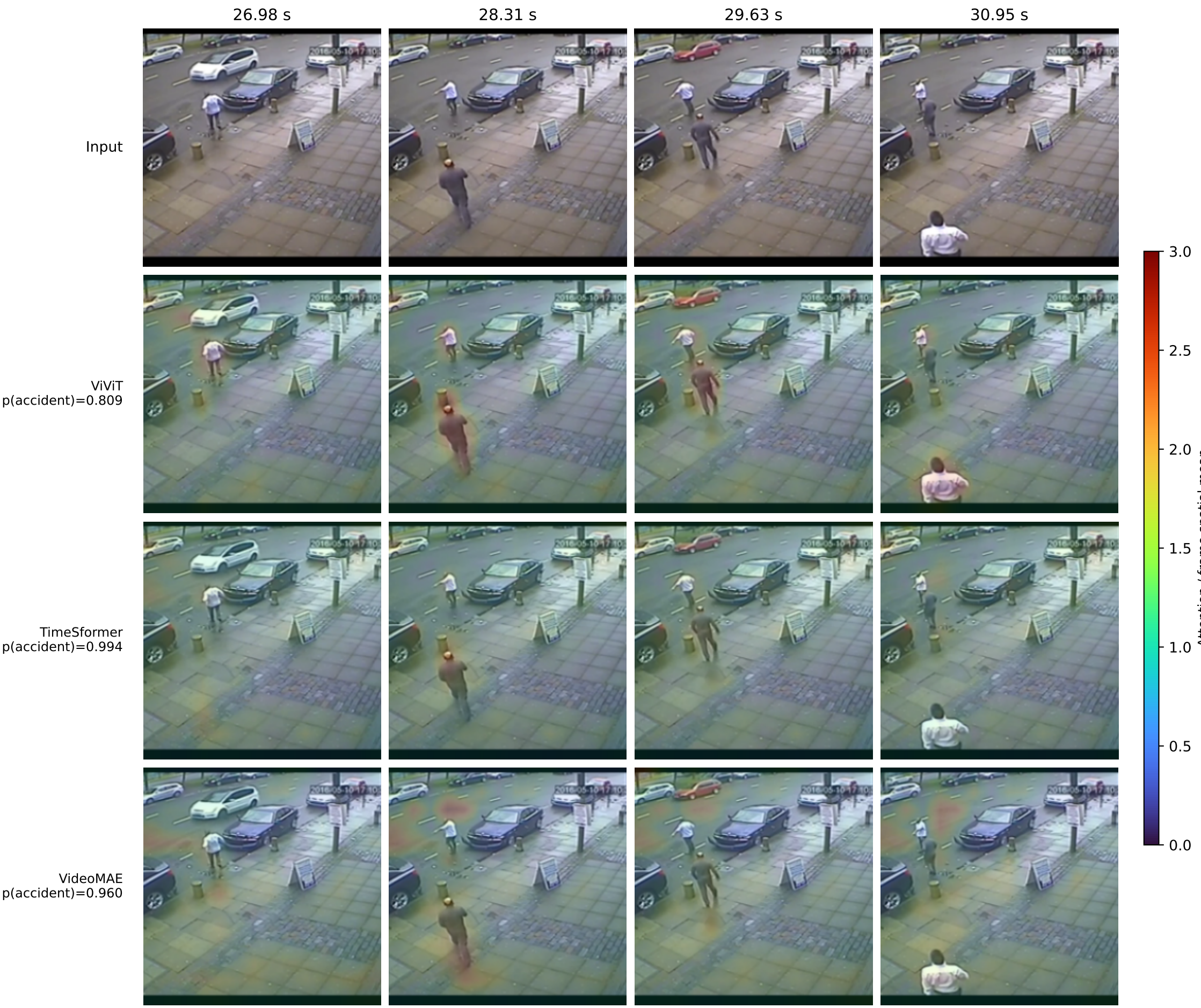
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A08 | Attention rollout | Accident label

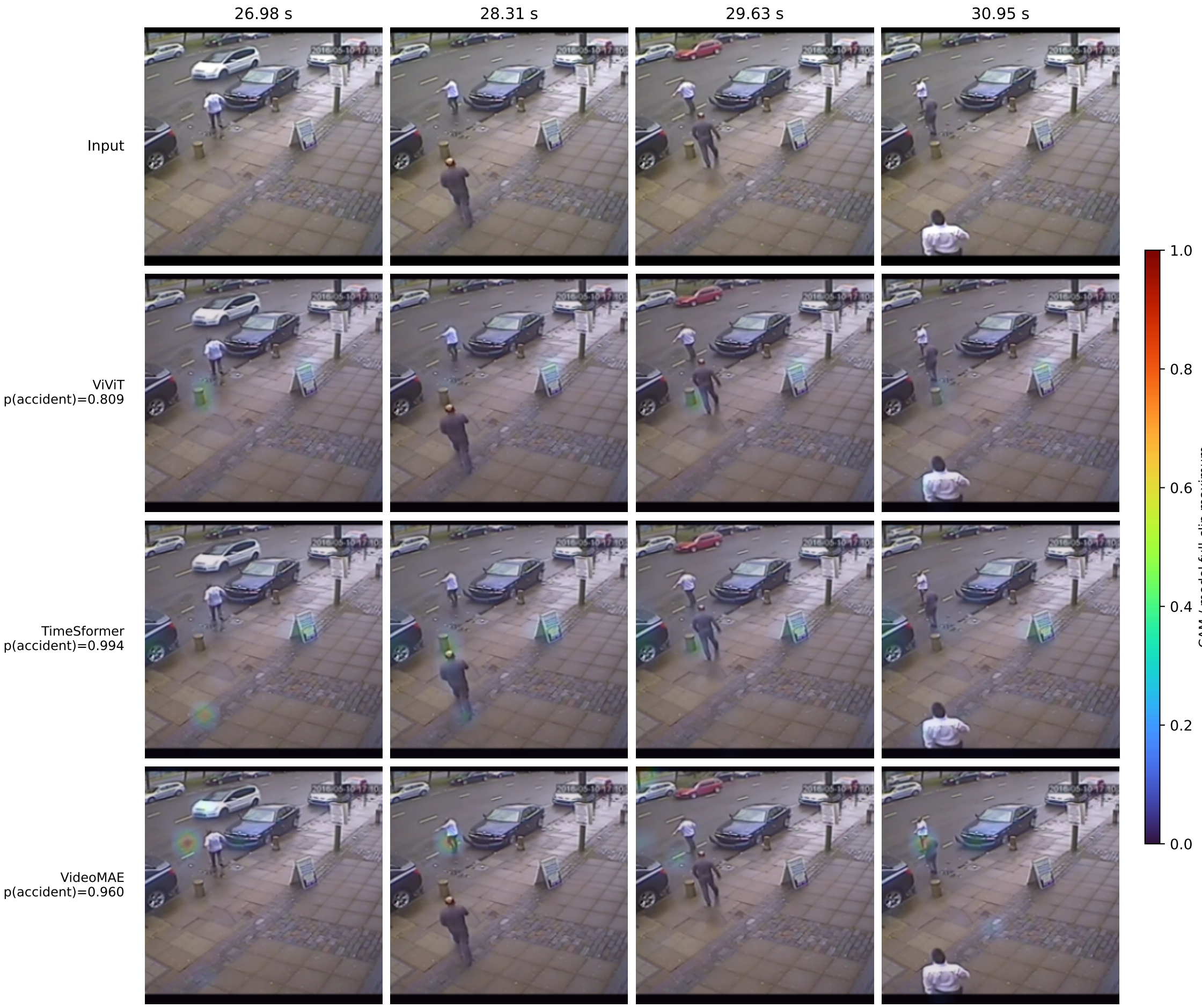
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A08 | LayerCAM | Accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

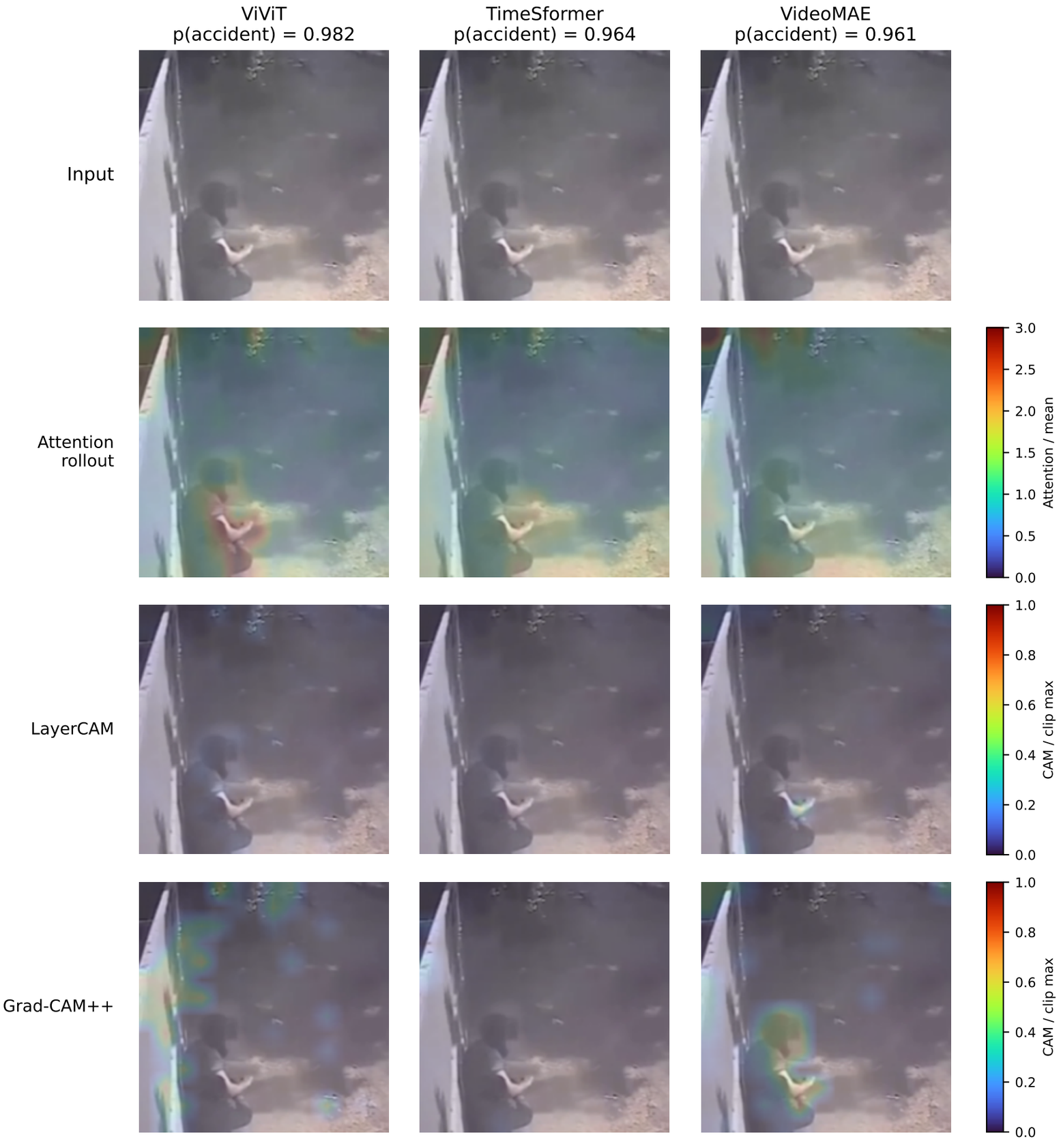
A08 | Grad-CAM++ | Accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A09 | Accident (dataset label) | t = 22.56 s

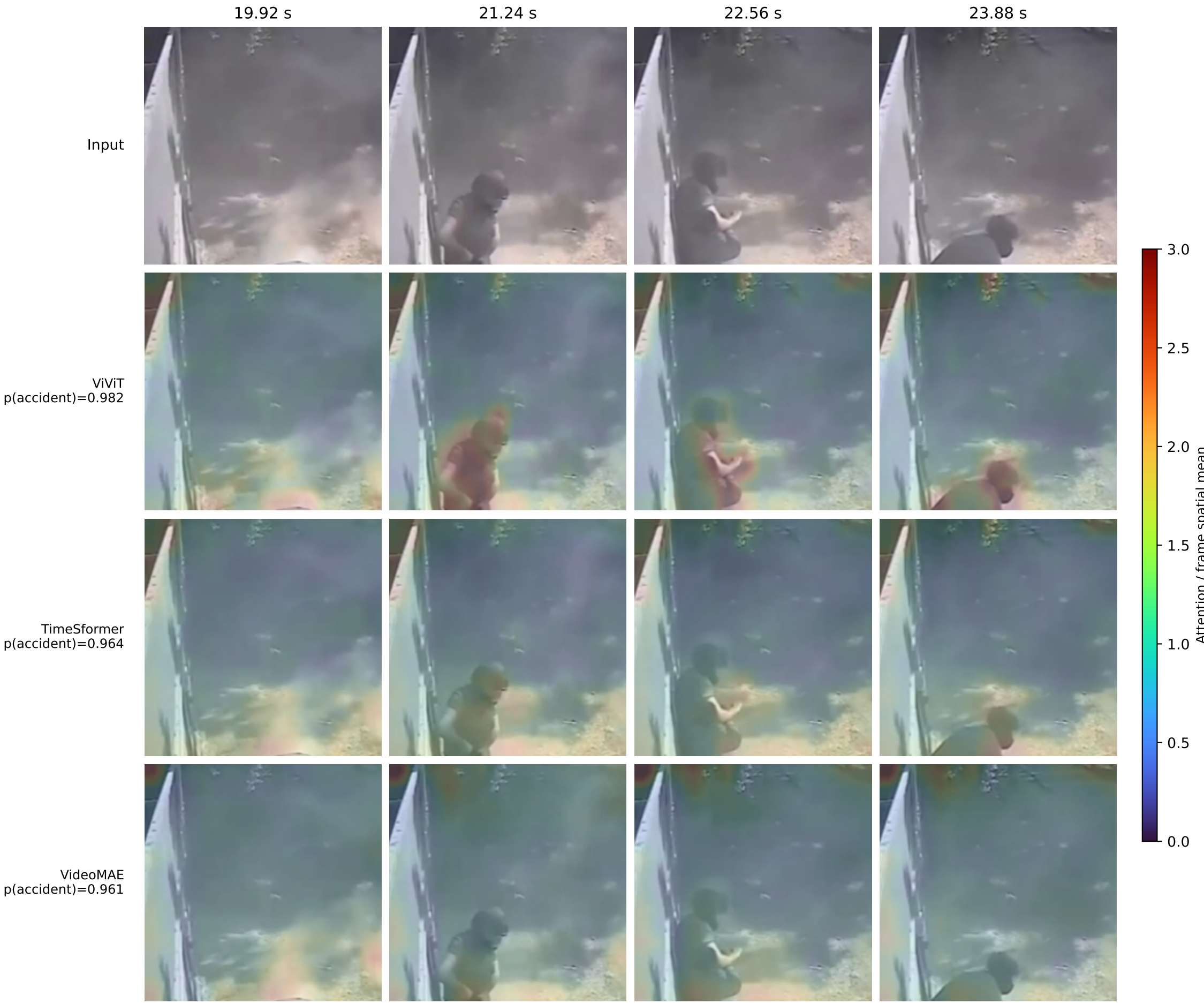
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A09 | Attention rollout | Accident label

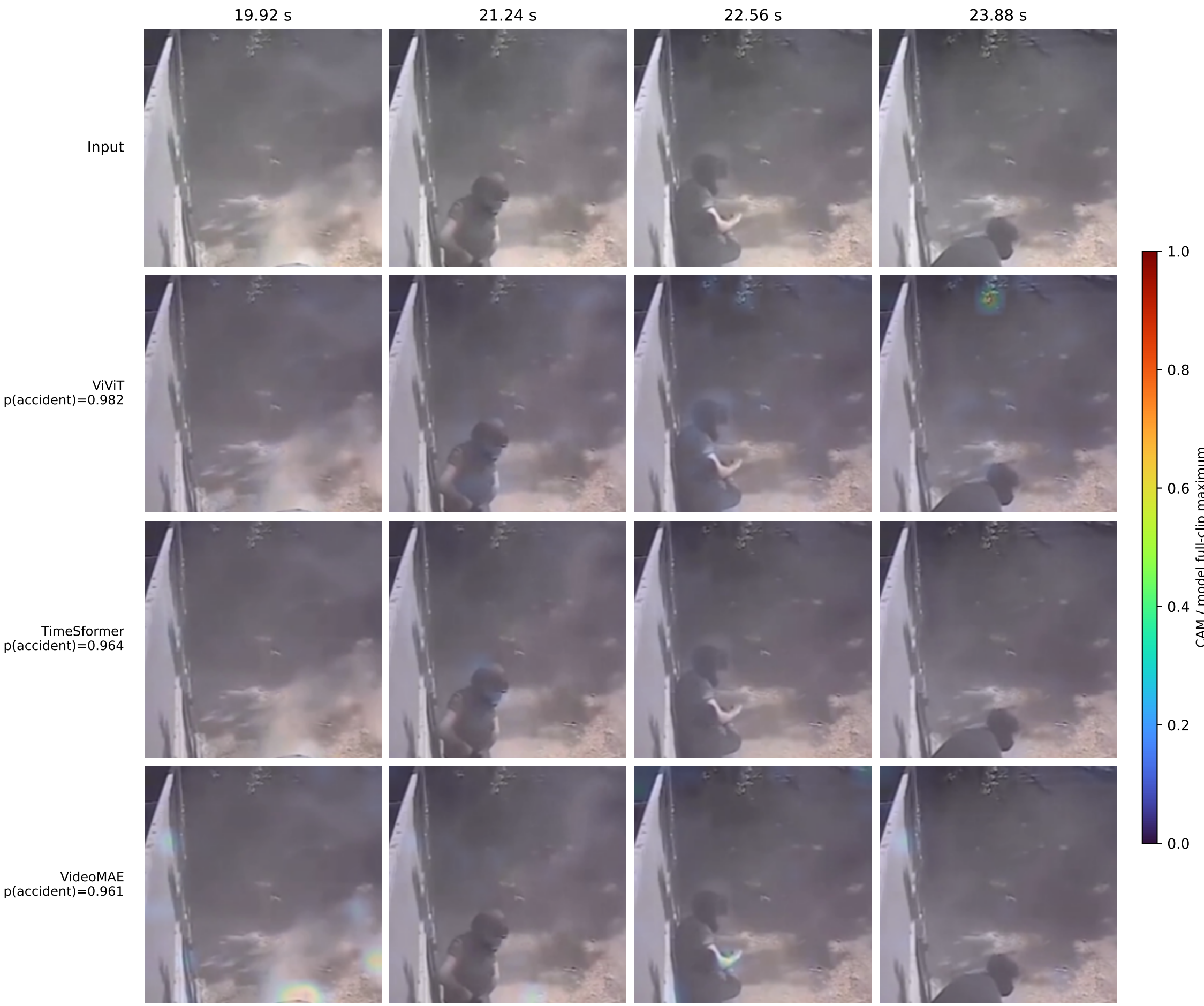
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A09 | LayerCAM | Accident label

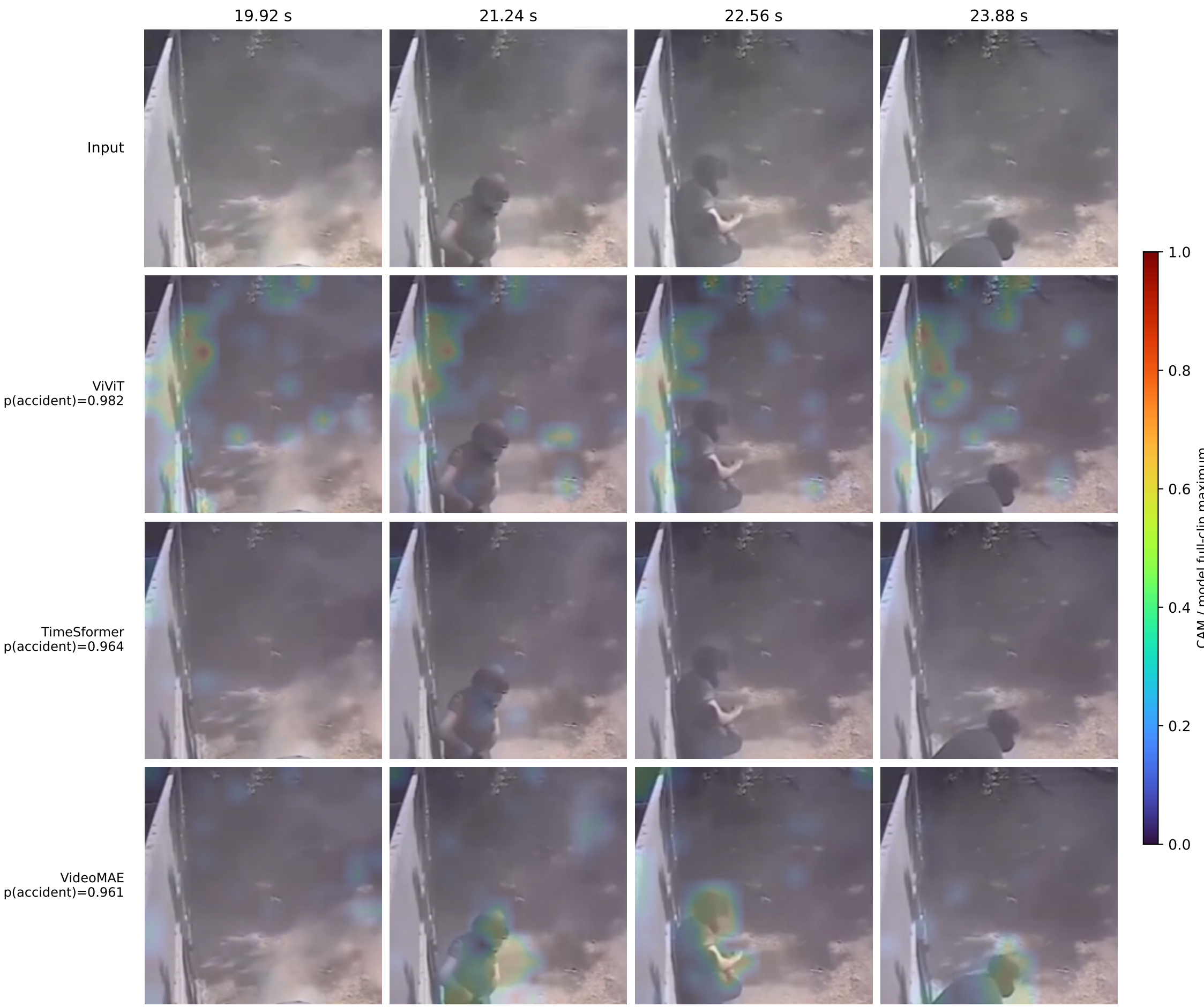
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A09 | Grad-CAM++ | Accident label

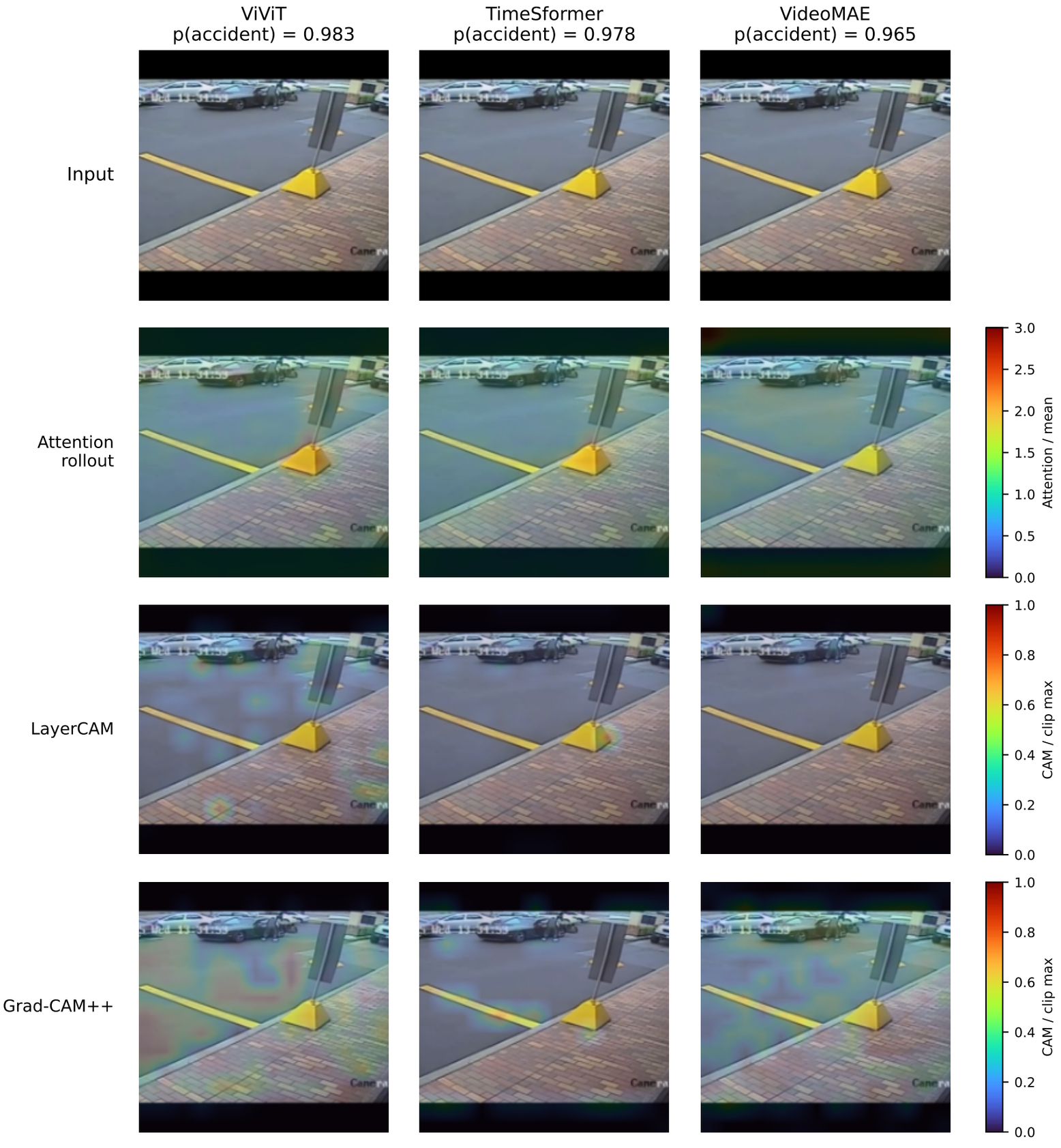
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A10 | Accident (dataset label) | t = 37.13 s

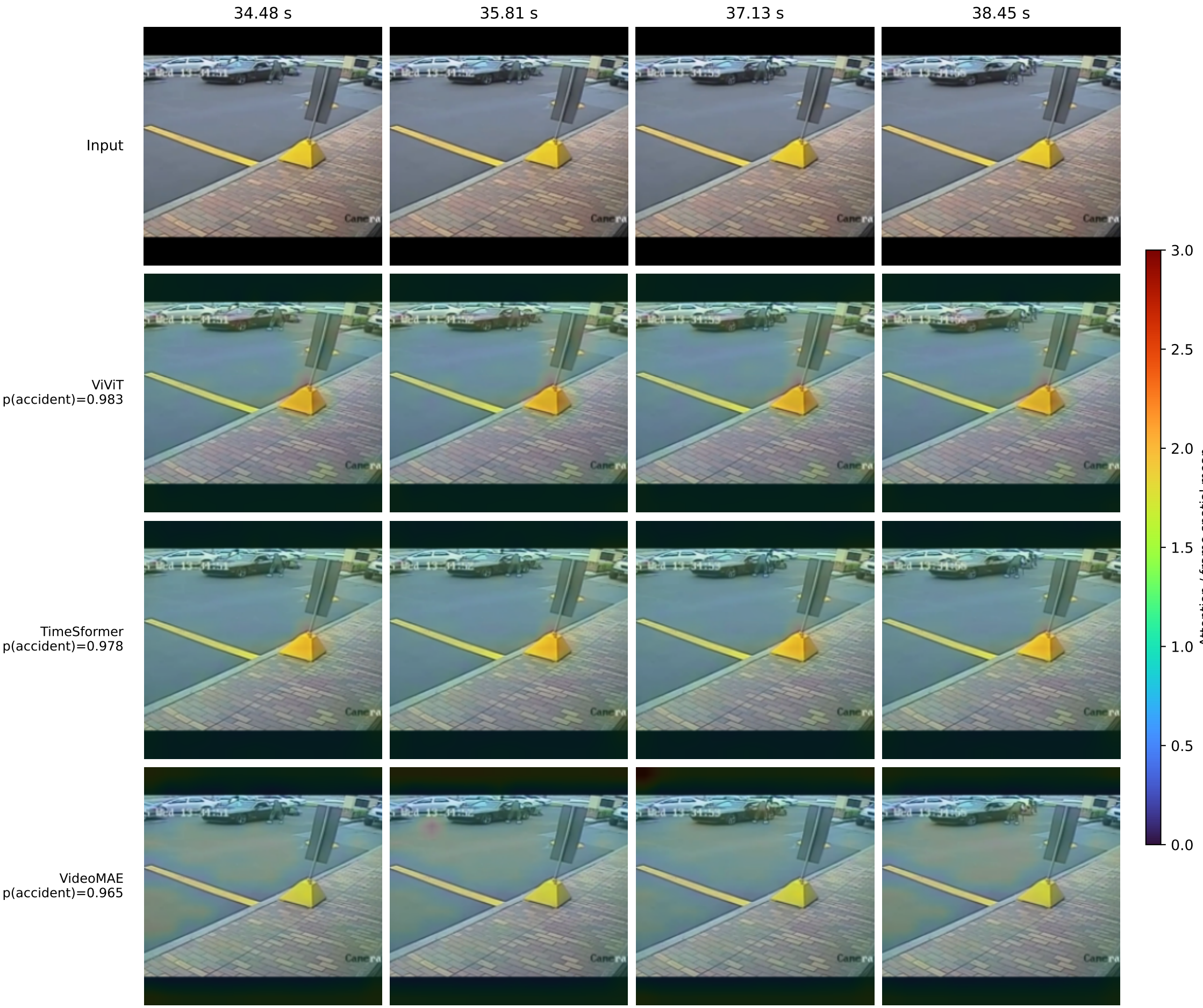
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

A10 | Attention rollout | Accident label

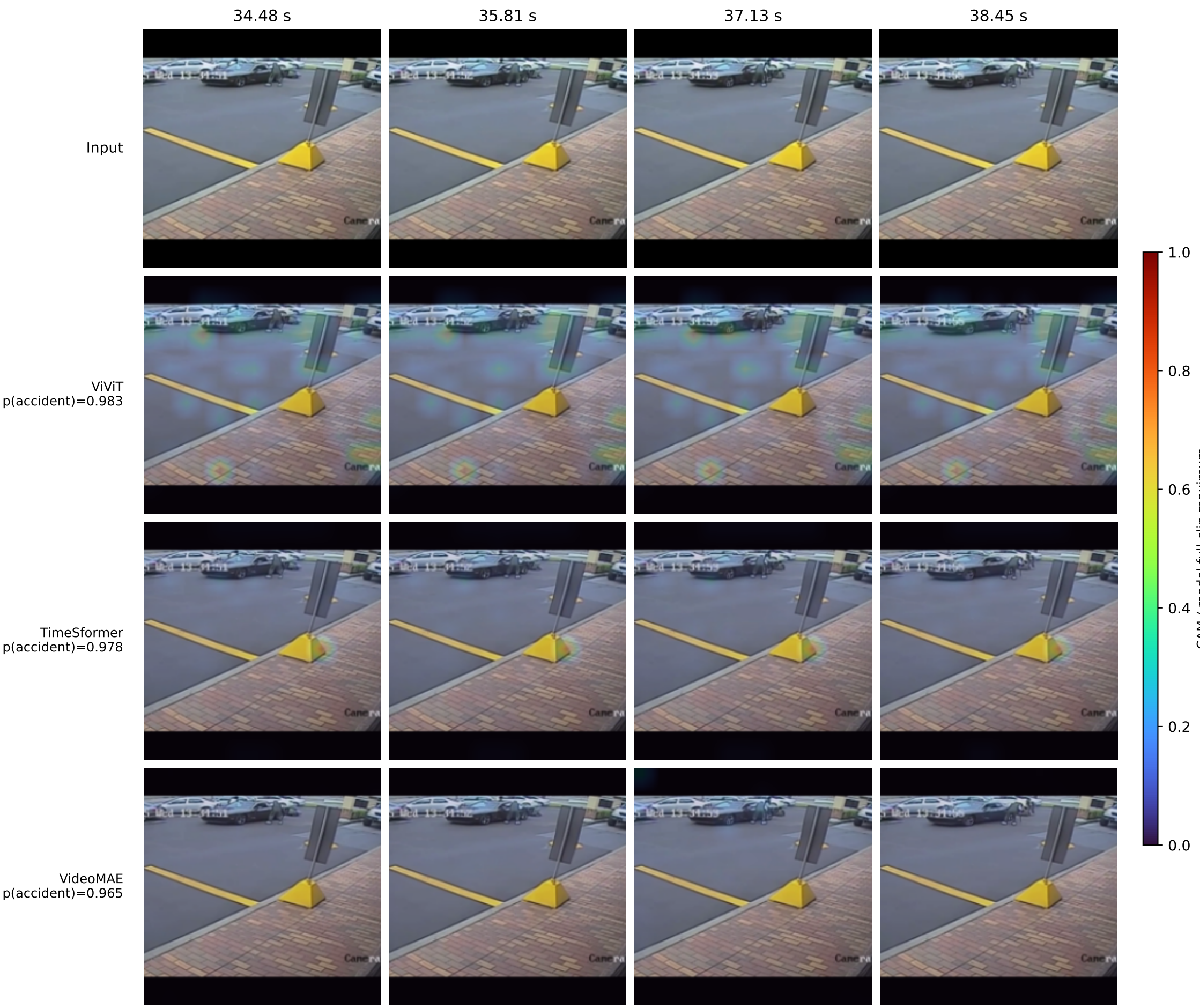
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A10 | LayerCAM | Accident label

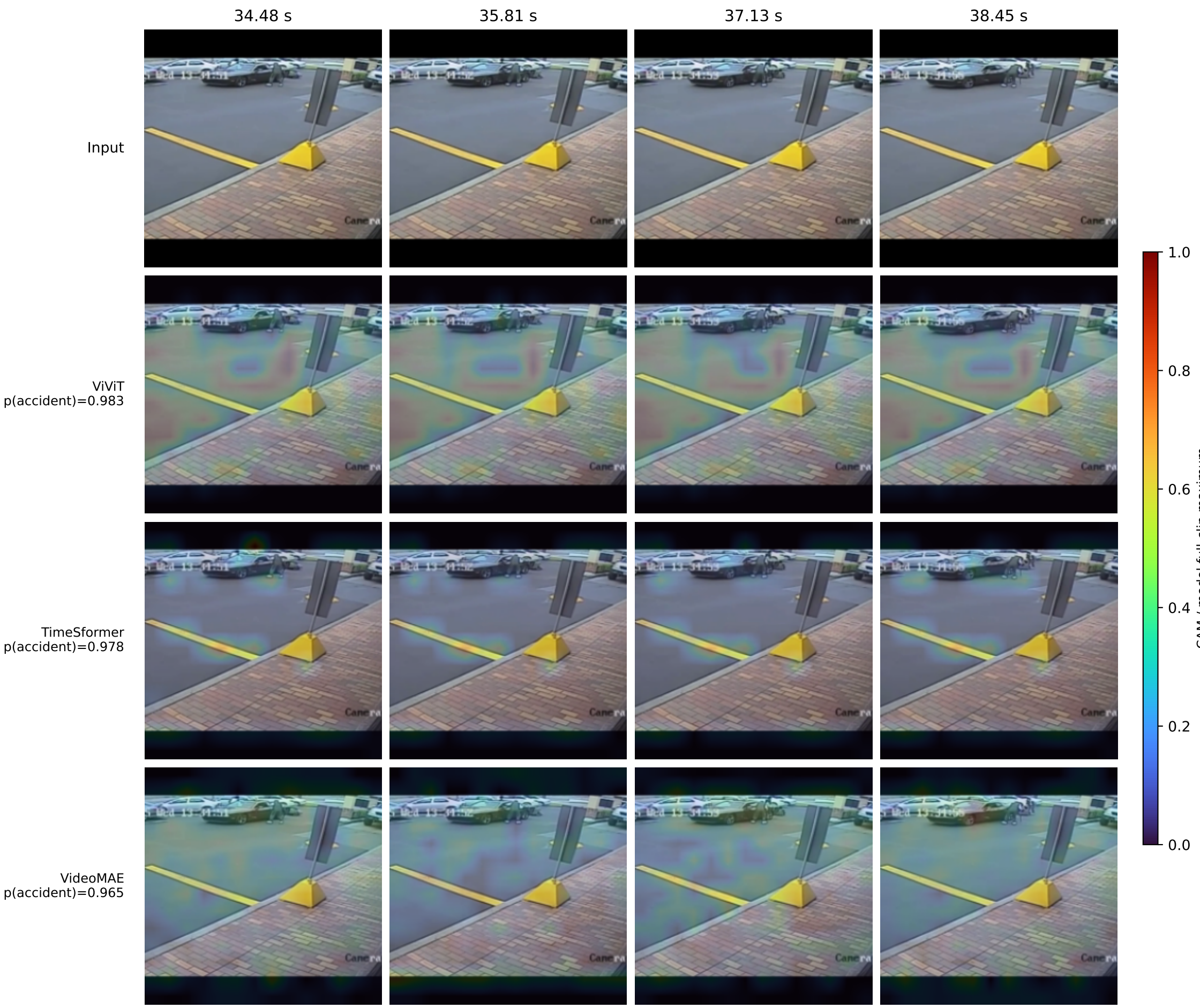
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

A10 | Grad-CAM++ | Accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.